

# Global transport in soaring flights

This deck develops the whole chapter without a prescribed duration. Its central distinction is between measured constraints on the observed ground trajectories and hypotheses about their generating process.

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Chapter 3

Thesis §3.1

## Displacement and time scales

$$\mathbf{X}_f(t, \tau) = \mathbf{r}_f(t + \tau) - \mathbf{r}_f(t), \quad R_f = |\mathbf{X}_f|.$$

$$M_2^{\text{flight}}(\tau) = \frac{1}{N_f(\tau)} \sum_f \underbrace{\frac{1}{n_f(\tau)} \sum_k R_{fk}^2}_{\text{mean within flight } f}.$$

Windows stay within a retained segment. The main comparison covers  $10 \leq \tau \leq 10^4$  s.

Displacement growth, distribution shape, directional preference and memory must be confronted together.

Archive TAMSD: equal segment weights. Full  $V_1, V_2$ : equal flight weights. Baseline quantiles and moments: pooled windows.

## The motion a stochastic model must reproduce

**Discussion.** What motion should a stochastic model reproduce?

The full diagnostic collector has no flight cap and no longest-segment-only restriction. Flights recorded more slowly than ten seconds are ineligible for its common grid. Each lag uses every supported flight; a separate fixed long-flight comparison controls population change. Different grids, minimum segment sizes and weights explain some differences from the archive TAMSD.

| Estimator       | Unit receiving equal weight           | Support requirement                     |
|-----------------|---------------------------------------|-----------------------------------------|
| Launch ensemble | available flights at elapsed time $t$ | observed coordinate at that time        |
| Archive TAMSD   | eligible retained segments            | native-grid displacement within segment |
| Full $V_1, V_2$ | eligible flights at lag $\tau$        | within-segment common-grid stencil      |
| Pooled moments  | available increment origins           | stated non-overlapping baseline stencil |

Different exponents can reflect different weights and changing support. Their difference alone is not an ergodicity test.

Launch, segment and flight averages use different weights

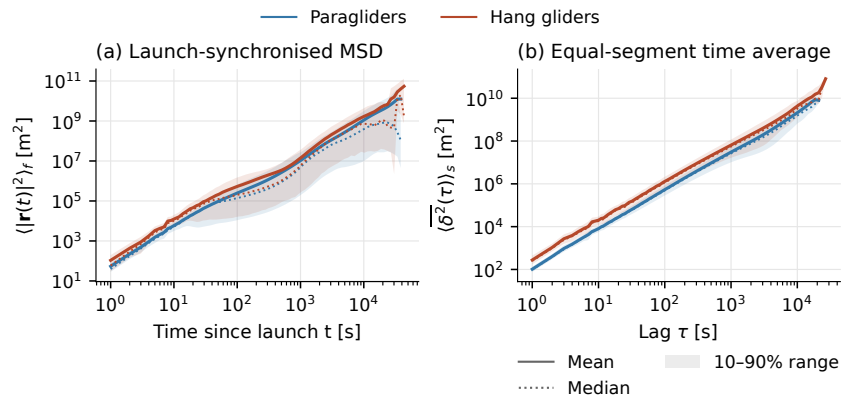
**Discussion.** Which estimator corresponds to the physical population the model should reproduce?

Within a flight, contributions are combined using their admissible origins. The archive TAMSD and the full diagnostic reporter use different grids and minimum segment sizes, so their flight counts need not be identical. State the estimator before comparing numerical slopes.

## Launch averages and time averages use different populations

**Discussion.** How much of the difference follows launch synchronization and segment weighting?

These are the upper two panels of the complete archive MSD figure. Blue denotes paragliders and terracotta hang gliders. The quantities use different clocks and weighting; their comparison is not an ergodicity test.



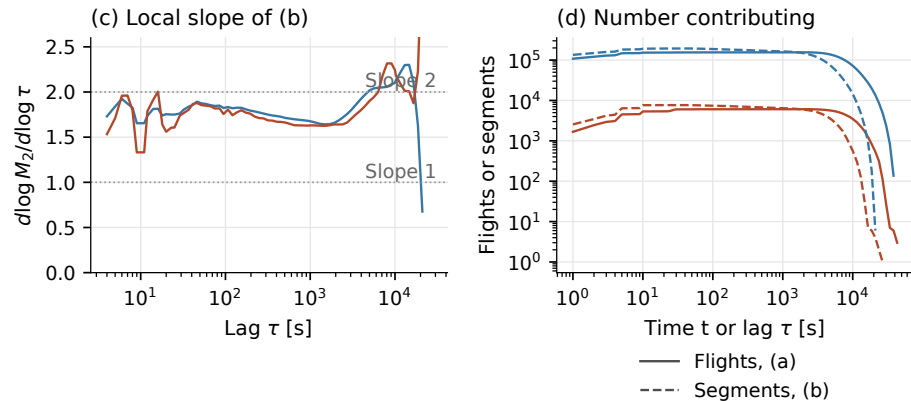
Thesis §3.1.1

## Effective and local growth exponents

**Local slopes must be read together with falling support**

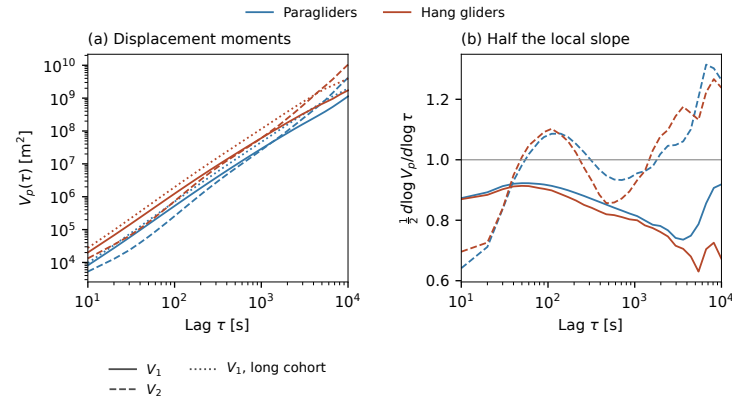
**Discussion.** Which scale interval supports a stable finite-range description?

The lower panels of the archive MSD figure show local slopes and contributing flights or segments. A small bootstrap error does not remove scale dependence or population selection.



Thesis §3.1.2

## Removing a constant velocity



Full eligible archive; 10–10,000 s. Solid:  $V_1$ ; dashed:  $V_2$ ; dotted:  $V_1$ , fixed long cohort. Right: half the local slope. Descriptive fits; no calibrated confidence intervals.

## The MSD grows faster than linearly on the measured interval

The displayed exponents are imported from the completed full report. The first- and second-difference estimators use equal flight weights and do not cross segment boundaries. Their local slopes bend, so these are descriptive fits over the stated range. They differ from the equal-segment archive TAMSD and from pooled-origin high-order moments.

|      | $b_1$ | $b_2$ |
|------|-------|-------|
| Para | 1.703 | 2.003 |
| Hang | 1.649 | 1.985 |

$V_1 = \langle R^2 \rangle$ ;  $V_2$  uses second differences.

The local slope bends. One fitted number does not establish a power law.

$$A_2 \mathbf{r} = \mathbf{r}(t + 2\tau) - 2\mathbf{r}(t + \tau) + \mathbf{r}(t), \quad V_2 = \langle |A_2 \mathbf{r}|^2 \rangle.$$

### Exact cancellation

$$\mathbf{r}(t) = \mathbf{r}_0 + \mathbf{v}_d t + \mathbf{Y}(t) \quad \Rightarrow \quad A_2 \mathbf{r} = A_2 \mathbf{Y}.$$

The constant velocity may combine wind, route choice and net progress. A turn or spatially varying wind survives.

At  $\tau = 10^4$  s,  $V_2$  retains only 14,360 para / 563 hang flights; it needs  $2\tau$  of continuous support.

### Conditional interpretation

If centred increments have finite variance, stationary statistics and the law  $K\tau^{2H}$ ,

$$V_2 = K(4 - 2^{2H})\tau^{2H}, \quad 0 < H < 1.$$

At  $H = 1$  the prefactor vanishes. Our  $b_2 \simeq 2$  cannot be read as a validated  $H \simeq 1$ .

## Constant velocity cancels; an intrinsic $H$ remains open

**Discussion.** Which residual and which scale interval could satisfy the assumptions needed to estimate  $H$ ?

Invariant means unchanged after adding a constant position offset and a constant velocity times time. It does not mean constant as  $\tau$  changes. For adjacent increments  $I_1$  and  $I_2$ , their sum spans  $2\tau$  and their difference is  $A_2$ . The parallelogram identity gives  $V_2 = 4M(\tau) - M(2\tau)$  for the same stationary-increment population. Substituting  $M(\tau) = K\tau$  to the power  $2H$  gives the displayed formula. All three conditions matter: finite variance, stationary increment statistics, and the same power law through  $2\tau$ . Stationarity means that the increment law at a given duration does not change with the start time; it does not require independence. A fitted  $b_2$  above two does not by itself identify which assumption fails. A random-phase circular trajectory has stationary increments and finite variance but no exact increment power law, and  $V_2$  grows as  $\tau$  to the fourth at small lag.

$$\bar{\mathbf{v}}_1 = \frac{\mathbf{r}(t + \tau) - \mathbf{r}(t)}{\tau}, \quad \bar{\mathbf{v}}_2 = \frac{\mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau)}{\tau}.$$

$$\frac{A_2 \mathbf{r}}{\tau} = \bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1, \quad D_2(\tau) = \frac{\sqrt{V_2(\tau)}}{\tau} = \sqrt{\langle |\bar{\mathbf{v}}_2 - \bar{\mathbf{v}}_1|^2 \rangle}.$$

$D_2$  is in m/s. At  $\tau = 60$  s it compares the mean velocity vector in one minute with the vector in the next minute. Changes of magnitude and direction both contribute.

Exact kinematic identity; no power law or stationarity assumption is needed.

## Second differences measure changes of interval-mean velocity

**Discussion.** At which scales does the ground motion change most, and how does that depend on the region?

This is an RMS change of interval-mean ground velocity. It is neither instantaneous acceleration nor wind speed. A constant added velocity cancels in both intervals. Keeping the physical units makes regional amplitudes interpretable even when a single exponent is not supported. The later regional study uses this quantity together with the vector covariance, distributions and order-three control.

$$V_2(\tau) = \tau^2 D_2^2(\tau), \quad b_{2,\text{loc}} = 2 + 2 \frac{d \log D_2}{d \log \tau}.$$

## Saved full-archive curves

Over 10–10,000 s,  $D_2$  varies within:

|              | $D_2$ range (m/s) |
|--------------|-------------------|
| Paragliders  | 4.9–7.3           |
| Hang gliders | 7.6–11.5          |

A nonmonotone change of less than a factor two across three decades of lag.

Thesis Sec. 3.1.2; saved full-archive variations, unchanged numerical run. Ranges are descriptive, not confidence intervals.

## Interpretation and limits

The  $\tau^2$  factor can dominate growth; local curvature prevents one validated  $H$ .

The fixed long-flight cohort also gives  $b_2 > 2$ : flight membership alone cannot explain it. Origins and weights still change.

## Why a slope near two need not identify a Hurst exponent

**Discussion.** Does a stable slope survive with flights and origins fixed, within comparable flight phases?

$D_2$  is the RMS difference between mean ground-velocity vectors in adjacent intervals, with the same per-lag averaging as  $V_2$ . These ranges were checked directly from the saved full-archive arrays; no flight subsampling or new transport fit was used. Approximate constancy explains how a descriptive slope near two arises, but does not identify its physical cause. At the exact  $H=1$  boundary of the stationary-increment power-law model, the second difference would have zero variance. Small fitted excesses above two have no calibrated significance here. Keeping only long flights fixes membership but not the contributing origins or their weights. Cadence and smoothing sensitivity should accompany a search for a restricted stable regime; the current evidence does not rule out every useful finite-range  $H$ .

A stationary centred velocity with variance  $\sigma_v^2$  and normalised correlation  $C(s) = e^{-s/t_p}$  gives

$$V_2(\tau) = 2\sigma_v^2 t_p^2 \left[ 2\frac{\tau}{t_p} - 3 + 4e^{-\tau/t_p} - e^{-2\tau/t_p} \right].$$

$$\tau \ll t_p : \quad V_2 \sim \frac{4\sigma_v^2}{3t_p} \tau^3, \quad \tau \gg t_p : \quad V_2 \sim 4\sigma_v^2 t_p \tau.$$

A local slope near two occurs between these regimes despite stationary position increments. [This is an analytical example, not a fit to the flights.](#)

In flights, turns, glide–climb transitions and changing wind survive constant-velocity cancellation. Their characteristic times, combined with changing observation support, are a plausible origin of the measured curvature.

Thesis Sec. 3.1.2 and the velocity-covariance model in Sec. 3.6; the closed formula was checked against covariance integration.

## Finite persistence can produce a crossover without nonstationarity

**Discussion.** Which dominates the bends: turning, phase changes or sampling? Can phase-conditioned and common-origin comparisons separate them?

For the same stationary-increment population,  $V_2(\tau) = 4M_2(\tau) - M_2(2\tau)$ . A 2H interpretation requires one increment power law at both  $\tau$  and  $2\tau$ , not merely stationarity. Exponential velocity covariance introduces a persistence time and gives a cubic-to-linear crossover; a differentiable curved path can instead give a quartic small-lag limit. Neither limit is claimed to describe all measured lags. The observed D2 curves are nonmonotone and need not follow one exponential model. Anisotropy alone does not invalidate the vector variance identity. We have a plausible explanation in terms of finite physical scales and sampling, not an identified wind effect or demonstrated nonstationarity. When conditioning on phases, retain an explicit category for windows crossing transitions.

Thesis §3.1.2

## Third differences and the distribution of changes

$$A_3 \mathbf{r} = \mathbf{r}(t + 3\tau) - 3\mathbf{r}(t + 2\tau) + 3\mathbf{r}(t + \tau) - \mathbf{r}(t), \quad V_3 = \langle |A_3 \mathbf{r}|^2 \rangle.$$

For  $\mathbf{r} = \mathbf{r}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2$  with constant  $\mathbf{a}$ ,

$$V_2 = |\mathbf{a}|^2 \tau^4, \quad V_3 = 0.$$

Under the centred stationary-increment power law,

$$\frac{V_3}{V_2} = \frac{15 - 6 \cdot 2^{2H} + 3^{2H}}{4 - 2^{2H}}, \quad 0 < H < 1.$$

Order three requires  $3\tau$  of continuous support; the model law must hold through  $3\tau$ .

### Third differences also cancel a constant acceleration

**Discussion.** Do the same supported observations satisfy the predicted relation between orders?

$V_3$  is a sensitivity check to quadratic position trends. A smooth turn survives: typically  $V_2$  scales as  $\tau$  to the fourth and  $V_3$  as  $\tau$  to the sixth at small lag. Under the stated model both orders have exponent  $2H$  but different amplitudes; Brownian motion plus constant velocity gives  $V_3/V_2=3$ . Independent position error with variance  $\sigma^2$  per coordinate contributes  $6\sigma^2$  at order two and  $20\sigma^2$  at order three. Correlated errors and smoothing require a different noise calculation. Agreement between orders does not prove that environmental forcing has been removed.

Thesis §3.1.3

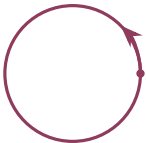
## Open and closed routes

Open and closed tasks change the growth of displacements

**Discussion.** Compare tasks at matched duration and region, using both physical lag and  $\tau/T$ .

The task declaration supplies open/closed labels; it does not force every retained segment to return to its start. Counts and slopes are read from the full-run report. Match duration and environment before interpreting task contrasts. The exact circular example illustrates how geometry survives second differencing; it is not a fitted flight model.

Open straight course



Closed circular course

Full eligible archive

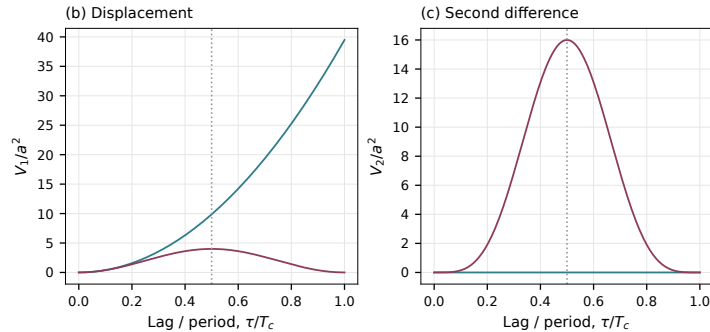
|      | Task   | <i>N</i> | <i>b</i> <sub>1</sub> | <i>b</i> <sub>2</sub> |
|------|--------|----------|-----------------------|-----------------------|
| Para | Open   | 67003    | 1.770                 | 1.937                 |
|      | Closed | 87807    | 1.623                 | 2.036                 |
| Hang | Open   | 2183     | 1.688                 | 1.892                 |
|      | Closed | 3877     | 1.628                 | 2.007                 |

The ordering reverses at order two. Constant-drift cancellation leaves the return geometry in the observable.

Task declarations; descriptive slopes, 10–10,000 s. Drawing is a geometric illustration. Exact circular example.

For  $\mathbf{r}(t) = a(\cos \omega t, \sin \omega t)$  and  $T_c = 2\pi/\omega$ ,

$$V_1 = 4a^2 \sin^2(\pi\tau/T_c), \quad V_2 = 16a^2 \sin^4(\pi\tau/T_c).$$



## A circle retains its geometry after second differencing

**Discussion.** Which scale effects could follow return geometry without a change of stochastic law?

This is an exact periodically continued circle, not a fitted flight model. Constant speed does not imply ballistic endpoint displacement over every lag. The original two variation panels are retained; the circular path was illustrated on the preceding task slide.

$$c_f^{\text{obs}} = \frac{|\mathbf{r}_{f,\text{last}} - \mathbf{r}_{f,\text{first}}|}{\sum_{s \in \mathcal{S}_f} \sum_k |\mathbf{r}_{fs,k+1} - \mathbf{r}_{fs,k}|}.$$

The denominator omits unobserved intervals. For a segmented flight,  $c_f^{\text{obs}}$  need not be bounded by one.

Declared task, observed endpoint separation and actual flown-route closure are related descriptions with different information requirements.

## Observed-path closure needs special care for segmented flights

**Discussion.** Does the task contrast persist when restricted to continuously observed flights?

The descriptor preserves recorded endpoints without inventing missing path length. Reacquisition offsets can affect it even though every increment used for transport remains inside one segment. Use the task labels and this observed-path ratio as distinct descriptive stratifications; neither is an independent route-quality label.

Thesis §3.2.1

## The scaling prediction

With  $a_\tau = (\tau/\tau_0)^H$ , the vector law  $\mathbf{X}(t, \tau) \stackrel{d}{=} a_\tau \mathbf{X}(t, \tau_0)$  implies

$$X_E(t, \tau) \stackrel{d}{=} a_\tau X_E(t, \tau_0), \quad X_N(t, \tau) \stackrel{d}{=} a_\tau X_N(t, \tau_0).$$

### Signed marginal densities

$$p_E(x; \tau) = a_\tau^{-1} f_E(x/a_\tau),$$

$$p_N(y; \tau) = a_\tau^{-1} f_N(y/a_\tau).$$

The exponent is shared; the shapes  $f_E$  and  $f_N$  can differ. No independence or isotropy is assumed.

### Magnitudes used for quantile fits

For  $j \in \{E, N\}$ ,

$$|X_j(\tau)| \stackrel{d}{=} a_\tau |X_j(\tau_0)|,$$

$$Q_{|X_j|,p}(\tau) = a_\tau Q_{|X_j|,p}(\tau_0).$$

Positive finite quantiles allow log fits.

Negative or zero signed quantiles do not.

Thesis Sec. 3.2.1. One-dimensional densities have factor  $a_\tau^{-1}$ ; the two-dimensional joint density has  $a_\tau^{-2}$ .

## Scalar vector scaling also constrains east and north

**Discussion.** Do both marginal scales agree, and does their joint dependence scale with them?

The signed east marginal is obtained by integrating the joint reference density over north, and conversely for north. Multiplication by a positive scalar commutes with each projection, absolute value and radial norm. For positive  $y$ , the absolute-component density is the sum of the signed density at  $y$  and minus  $y$ . If only separate marginal scaling is assumed, each component may instead have its own exponent  $H_E$  or  $H_N$ . This weaker hypothesis does not determine the joint law, even if the two exponents coincide. The later east, north and radial quantile plots concern magnitudes rather than signed component quantiles; direction and dependence are retained in the subsequent signed joint maps.

$$Q_p(\tau) = \inf\{r : F_\tau(r) \geq p\}.$$

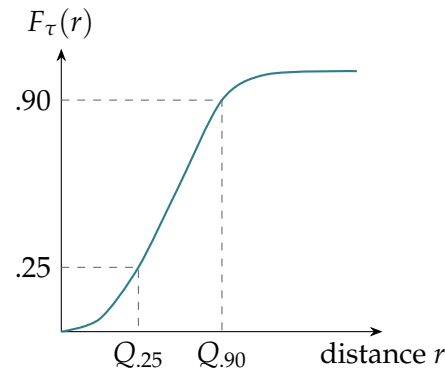
An  $H$ -self-similar process with stationary increments predicts

$$R(\tau) \stackrel{d}{=} (\tau/\tau_0)^H R(\tau_0),$$

$$Q_p(\tau) = (\tau/\tau_0)^H Q_p(\tau_0).$$

$$H_p = H \text{ for every } p; \quad Q_{.90}/Q_{.25} \text{ stays fixed.}$$

Schematic CDF. Quantiles describe ranks within one distribution; they do not define permanent classes of slow and fast flights. Equal slopes are necessary, not sufficient, for process self-similarity.



### A single displacement scale predicts equal quantile slopes

The lower quartile probes relatively small displacements, while the 90th percentile probes larger ones. Both use the same distribution at each lag. A different flight may supply an observation near each rank, and rank identities may change; that does not invalidate a quantile. The mathematical issue is whether the population and weighting used to construct the distribution change with lag. No isotropy assumption is needed for the radial scaling implication.

For each supported origin, measure the paired displacement  $\mathbf{X}(t, \tau) = \mathbf{r}(t + \tau) - \mathbf{r}(t)$ , then form

$$Y_E = |X_E|, \quad Y_N = |X_N|, \quad Y_R = \sqrt{X_E^2 + X_N^2}.$$

For each quantity  $j$ , sort the values at that lag and accumulate their weights:

$$\hat{F}_{j,\tau}(y) = \sum_{f,k} w_{fk}(\tau) \mathbf{1}\{Y_{jfk}(\tau) \leq y\}, \quad \sum_{f,k} w_{fk}(\tau) = 1.$$

Read the first observed value where cumulative weight reaches  $p = .25, .50, .75, .90$ . Repeat at every lag to obtain four curves  $\hat{Q}_{j,p}(\tau)$ .

**Pooled windows:** equal weight per window. **Fixed-population control:** equal total weight per flight, with common origins.

Thesis Sec. 3.2. Quantiles use inverse empirical CDFs without interpolation; flight-specific quantiles are not averaged.

## From displacement windows to one quantile at each lag

**Discussion.** Which population and weights define the distribution whose quantiles we are fitting?

Each lag supplies a new distribution of displacement magnitudes, not a list of permanent slow and fast flight classes. The baseline pools non-overlapping supported windows from all eligible flights at that lag. With equal window weight, the quantile is the observation of rank ceiling( $p$  times number of windows). The fully controlled analysis gives an origin in flight  $f$  weight  $1/(Nn_f)$ , keeping flights, origins and weights fixed across lags. Its origins overlap and are not independent. All coordinates use the same paired windows; the radial quantile must be computed from each window's radius, not from the component quantiles. No drift subtraction is applied to these quantile curves.

For one discipline, quantity  $j$  and probability  $p$ , set

$$x_\ell = \log \frac{\tau_\ell}{1000 \text{ s}}, \quad y_\ell = \log \left[ \hat{Q}_{j,p}(\tau_\ell) / (1 \text{ m}) \right].$$

Fit a straight line with a free intercept by ordinary least squares:

$$(\hat{b}_{j,p}, \hat{H}_{j,p}) = \arg \min_{b,H} \sum_{\ell \in \mathcal{L}} (y_\ell - b - Hx_\ell)^2, \quad \hat{Q}_{j,p}^{\text{fit}}(\tau) = (1 \text{ m}) e^{\hat{b}_{j,p}} \left( \frac{\tau}{1000 \text{ s}} \right)^{\hat{H}_{j,p}}.$$

- Full interval: 31 saved lags from 10 to 10,000 s, equal weight per lag.
- Repeat for three quantities and four ranks: [12 slopes per discipline](#).
- A fitted slope summarises the curve; curvature can remain around the line.

Thesis Secs. 3.2–3.3. Only positive supported quantiles enter log fits. The time reference changes the intercept, not the slope.

## Each fitted $H_{j,p}$ is the slope of one log–log quantile curve

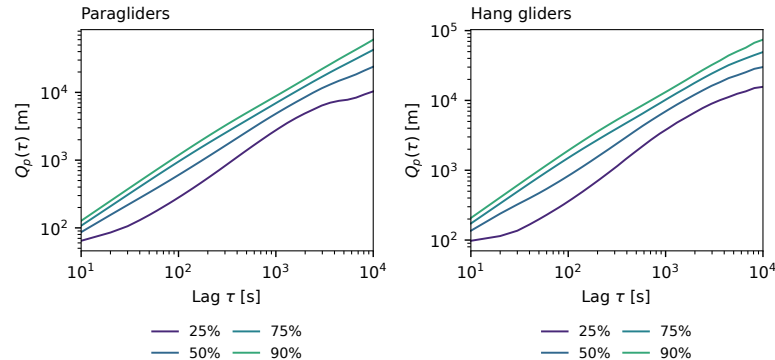
**Discussion.** Does one straight line describe these lags, and does its slope depend on rank or direction?

These are regressions of log quantile against log lag, not quantile regression of individual displacements, not fits to histogram density, and not slopes read off an exponent-versus-percentile plot. Natural and base-ten logarithms give the same exponent when used on both axes. The displayed 1000-s reference matches the fixed-population producer; the pooled producer uses seconds directly, which changes only the intercept. All 31 lags are positive and supported in the saved full-range results. When comparing the four sampling controls, the code uses their common positive lag set. The separate fixed-population fits are repeated on 60–1970 s and 210–1970 s for sensitivity. No inverse-variance weighting is used. These descriptive H values are not independently validated process Hurst exponents.

**Archive quantile curves are the input to the fits**

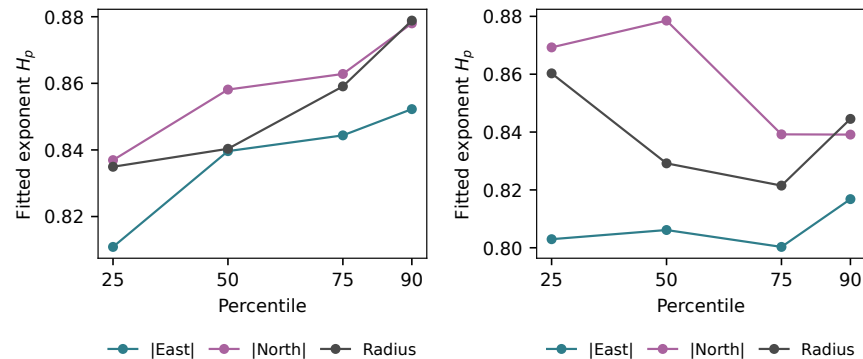
**Discussion.** Are the four log–log curves approximately parallel, and where do they bend?

This is a vector crop of the thesis figure, with its original curves, axes and legends. Each coloured curve contains one empirical radial quantile at every lag. Its log–log regression yields one radial  $H$  value in the next slide. The east and north curves are fitted by the same procedure, using absolute component magnitudes. The eligible population and number of windows can change with lag; this baseline precedes the fixed-population controls. These are empirical curves, not fitted lines.



Thesis Fig. 3.4, top row. Radial quantiles; all eligible windows pooled at each lag. Purple to green: 25th, 50th, 75th, 90th percentiles.

## Each point here summarises a separate quantile fit



Thesis Fig. 3.4, middle row. Left: paragliders; right: hang gliders. Fits span 10–10,000 s; the horizontal axis is percentile, not lag.

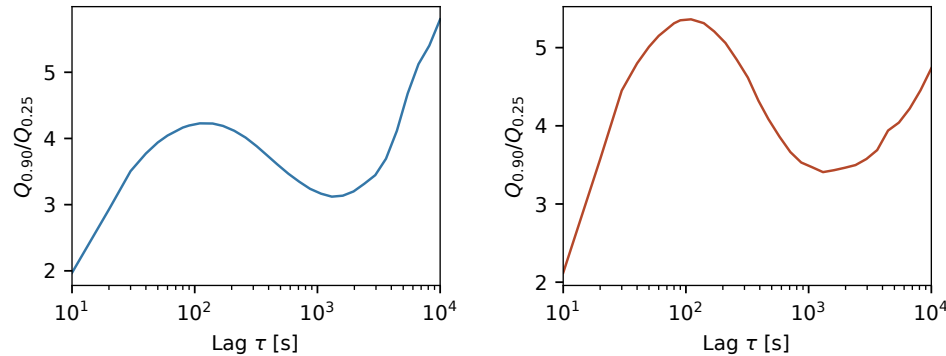
**Discussion.** Can the rank and directional differences be explained by changing flights or window weights?

For example, the radial point at percentile 50 is the slope of the median curve in the preceding slide. Repeat the regression separately for all four ranks and all three quantities to obtain this panel. Lines joining the points guide the eye; they are not another regression. These pooled-archive slopes have no calibrated confidence intervals. They differ from the later fixed-flight estimates because the empirical distributions use different flights, origins and weights. Rank ordering is therefore not yet evidence for an intrinsic transport mechanism.

**The archive quantile ratio also reveals changes with lag**

**Discussion.** Would this change in relative spread survive if the contributing population stayed fixed?

A common multiplicative displacement scale would cancel in this ratio and leave a constant. The measured ratio rises, falls and rises again, so a single fitted exponent difference compresses a curved evolution. This is the changing-pool baseline; the fixed-population ratio appears after the sampling controls and must not be substituted for it without explanation. Both distribution shape and population composition can contribute here.



Thesis Fig. 3.4, bottom row.  $Q_{.90}/Q_{.25}$  uses the same pooled radial curves as the first row. Left: paragliders; right: hang gliders.

Thesis §3.2.2

**A fixed population across lags**

| Population                              | Paragliders | Hang gliders |
|-----------------------------------------|-------------|--------------|
| Retained cleaned flights                | 156,406     | 6,094        |
| Eligible for the common-grid diagnostic | 155,085     | 6,060        |
| Fixed long-flight control               | 14,360      | 563          |

Every eligible segment contributes where it supports the lag. The long-flight control adds a common population and common origins across the complete lag range.

### The full archive and the fixed cohort answer different questions

**Discussion.** Which statement concerns the observed archive, and which is conditional on long flights?

The common ten-second grid requires native cadence at most ten seconds and sufficient within-segment support. These are eligibility conditions, not flight-number caps imposed by the Mac. The twenty-thousand-second criterion is an operational choice for the additional control. It is not mathematically necessary merely to observe one increment at a ten-thousand-second lag.

All eligible flights contribute at each supported lag in the archive-wide analysis. The additional fixed cohort contains every flight with a retained segment lasting at least 20,000 s.

This provides at least 1001 common ten-second origins in one qualifying segment at the largest lag of 10,000 s. Shorter segments of those flights contribute wherever they support the same full lag stencil.

### The fixed control requires common temporal support

**Discussion.** How different are the shorter-flight results on a shorter common lag interval?

The 20,000-second threshold is more restrictive than mere existence of a 10,000-second increment. It is an explicit support convention for this sensitivity analysis. The need for long observations is mathematical; the particular factor of two and chosen maximum lag are design choices, not necessities imposed by memory limits.

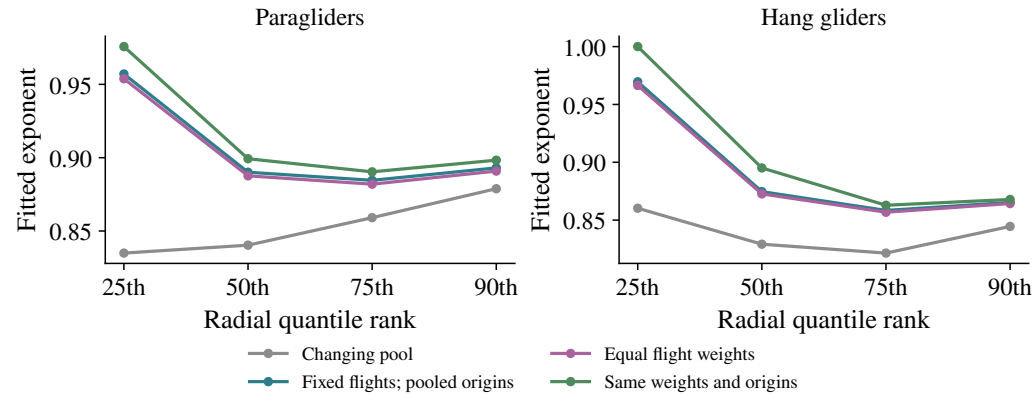
1. All available flights and origins, pooled.
2. Fixed flights, still pooling their available origins.
3. Fixed flights with equal total flight weight.
4. Fixed flights, equal flight weight and the same origins at every lag.

$$F_{\tau}(r) = \frac{1}{N_*} \sum_f \frac{1}{n_f^*} \sum_{k \in \mathcal{O}_f^*} \mathbf{1}\{R_{fk}(\tau) \leq r\}.$$

## Holding flights fixed does not by itself hold the observations fixed

**Discussion.** How much of a change in the quantile curve comes from membership, weights or origin selection?

All common origins stay within a retained segment for every requested lag. Overlapping increments are deliberately retained; they are not independent replicates. The final control uses 14360 paraglider flights with 21739115 common origins and 563 hang-glider flights. Read the final hang-glider origin count from its completed report rather than a preflight estimate.



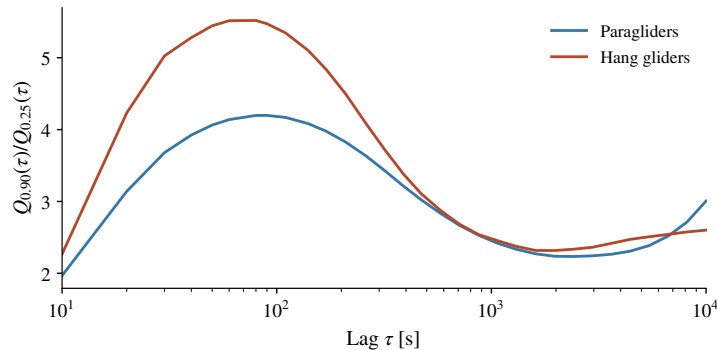
## Membership, flight weights and common origins change the comparison

**Discussion.** Which part of the slope change comes from each population control?

The four controls use the same ranks and fitting interval. The fixed cohort includes every qualifying long flight. Curves here are descriptive point estimates; the final fixed-control uncertainty uses the whole-flight paired bootstrap.

Thesis §3.2.3

## Relative spread



$$\frac{d \log(Q_{.90}/Q_{.25})}{d \log \tau} \\ = H_{.90}(\tau) - H_{.25}(\tau).$$

The ratio first rises, then falls, then rises again.

A negative fitted slope summarizes part of that evolution; it does not imply narrowing at every lag.

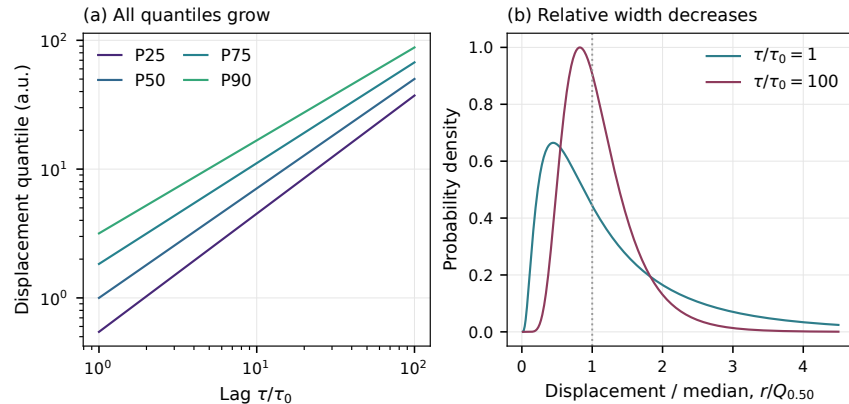
Fully controlled population. A lower ratio means less relative spread by this measure; it does not establish a flatter density tail.

## Relative spread changes, but the change is not monotone

**Discussion.** How much of this shape evolution survives site–day uncertainty, route matching and phase conditioning?

The fixed-control ratio starts near 1.97 for paragliders and 2.27 for hang gliders, peaks near ninety and eighty seconds, and has a post-peak minimum near two thousand seconds. The final values near 3.01 and 2.60 exceed the initial ones. A negative fitted log-ratio slope is therefore not monotone narrowing.

# All quantiles can grow while spread relative to the median narrows



## All quantiles can grow while spread relative to the median narrows

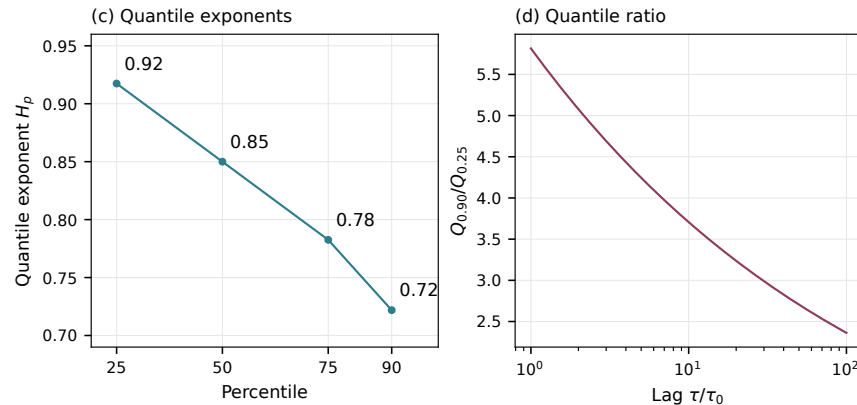
**Discussion.** How can a changing distribution shape coexist with increasing displacement scales?

Analytical illustration:  $R(\tau) = (\tau/\tau_0)^{0.85} \exp[\sigma(\tau)Z]$ , with standard Gaussian  $Z$  and  $\sigma(\tau) = 0.9 - 0.1 \log(\tau/\tau_0)$  on the illustrated range. This is not a fit to flight data.

## Unequal rank exponents can encode that changing shape

**Discussion.** Which empirical observations would distinguish this example from a true common scale family?

For the same lognormal illustration,  $H_p = 0.85 - 0.1 \Phi^{-1}(p)$ . The example explains the algebra without assigning permanent slow / fast identities to particular flights. The actual empirical curves need their own population controls and uncertainty estimates.



Thesis §3.3.1

## Testing east, north and radial scaling

$$\mathbf{X}(\tau) \stackrel{d}{=} \tau^H \mathbf{Z} \implies |\mathbf{X}(\tau)| \stackrel{d}{=} \tau^H |\mathbf{Z}|,$$

even if the law of  $\mathbf{Z}$  is anisotropic.

With two centred independent Gaussian components instead,

$$\mathbb{E}[R^2] = a\tau^{2H_E} + b\tau^{2H_N},$$

unequal component exponents can produce a lag-dependent radial effective exponent.

## Anisotropy does not prevent a common scalar rescaling

**Discussion.** Is a mismatch caused by unequal directional scales, changing marginal shape or changing joint structure?

Scalar self-similarity and isotropy are separate hypotheses. Operator self-similarity allows a matrix exponent and is broader, but does not permit arbitrary lag-by-lag normalization chosen to force collapse. The current plots assess the stated scalar comparison; they do not exhaust that wider model class.

$\{(1,2), (-1,-2)\}$       versus       $\{(1,-2), (-1,2)\},$

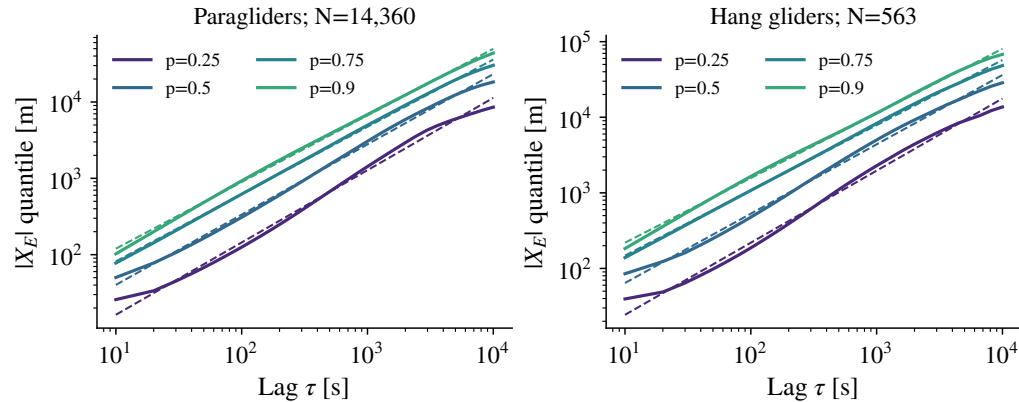
each pair equiprobable.

| Quantity              | First law  | Second law |
|-----------------------|------------|------------|
| Signed east marginal  | $\pm 1$    | $\pm 1$    |
| Signed north marginal | $\pm 2$    | $\pm 2$    |
| Radius                | $\sqrt{5}$ | $\sqrt{5}$ |
| Cross-covariance      | $+2$       | $-2$       |

The missing information is real: signs and component dependence require a joint diagnostic.

**East, north and radius together still do not identify the joint law**

This is an exact mathematical counterexample, not a simulated flight result. Even signed marginals and radius can agree while the joint laws differ. The Cramer–Wold characterization uses all signed linear projections, not only the two coordinate projections. The signed joint plots address additional structure at their stated bin resolution.

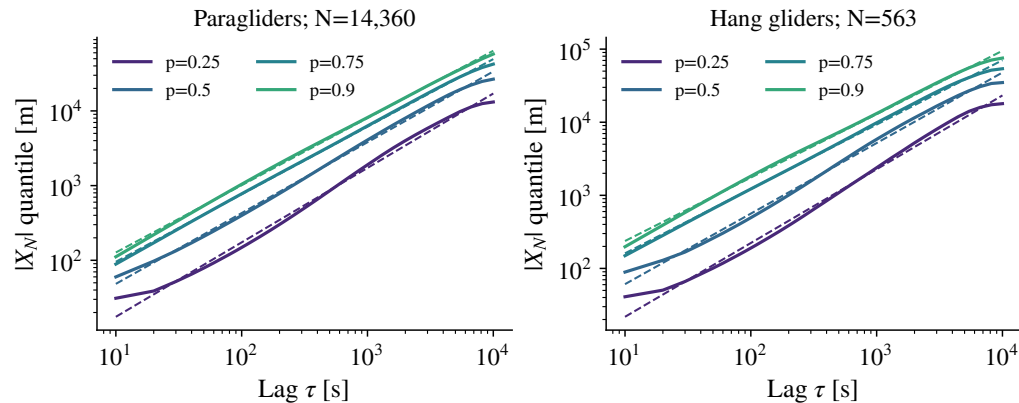


Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one  $H_{E,p}$ .

### Fixed-population quantiles: east component

**Discussion.** Do the ranks retain one relative shape as the lag increases?

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.

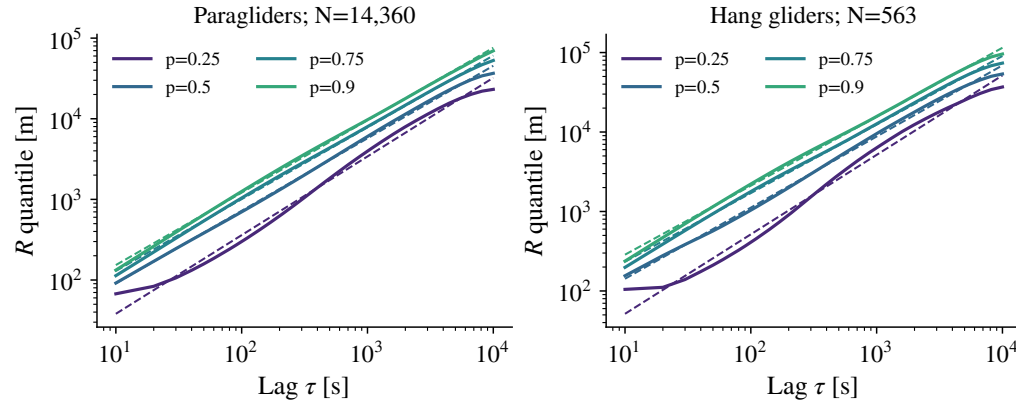


Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one  $H_{N,p}$ .

### Fixed-population quantiles: north component

**Discussion.** Do the ranks retain one relative shape as the lag increases?

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.



Solid: weighted empirical quantiles. Dashed: separate saved log-log fits over 10–10,000 s. Each dashed slope is one  $H_{R,p}$ .

### Fixed-population quantiles: radial displacement

**Discussion.** Do the ranks retain one relative shape as the lag increases?

Both disciplines use the same four ranks and their own complete fixed cohort. East and north use absolute signed-component magnitudes; radius is endpoint separation, not flown path length. Flights, origins and weights remain fixed across lags.

Resample flights with replacement 400 times. Each selected flight retains all its common origins and its joint east–north observations at every lag.

Recompute the weighted CDFs, extract all quantiles and repeat the same log–log fits in every draw. The 2.5th and 97.5th percentiles of the fitted slopes give the displayed 95% intervals; contrasts are computed within each draw.

Overlapping origins remain within the resampled flight; they are not treated as independent replicates.

### Whole-flight bootstrap preserves pairing across ranks, axes and lags

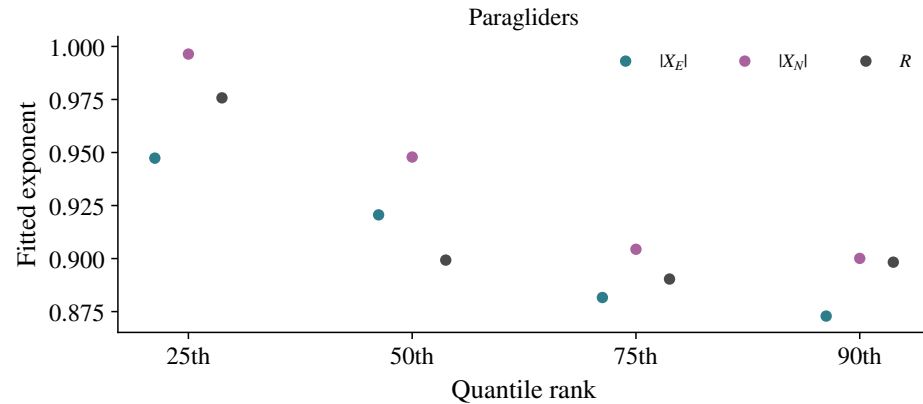
**Discussion.** How would shared pilot history or site–day weather change these uncertainty estimates?

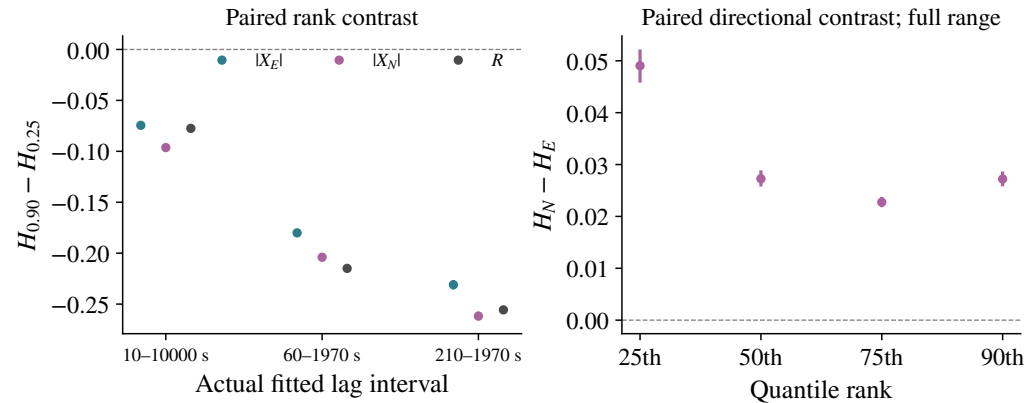
The current intervals assume independent flights within the fixed long-flight population. They are not iid increment intervals, simultaneous bands over all comparisons, or measurement-error intervals. Paired differences are computed inside each replicate. A large complete archive does not remove the assumptions behind the resampling model.

**Paragliders: component and radial exponent estimates**

**Discussion.** Which differences need a paired contrast instead of comparing overlapping intervals?

Markers are full-range fitted exponents; vertical segments are the exact saved 95-percent percentile endpoints from 400 whole-flight bootstrap draws. They are not simultaneous intervals and do not include sensor, site-day or fitting-range uncertainty.



**Paragliders: paired rank and directional contrasts**

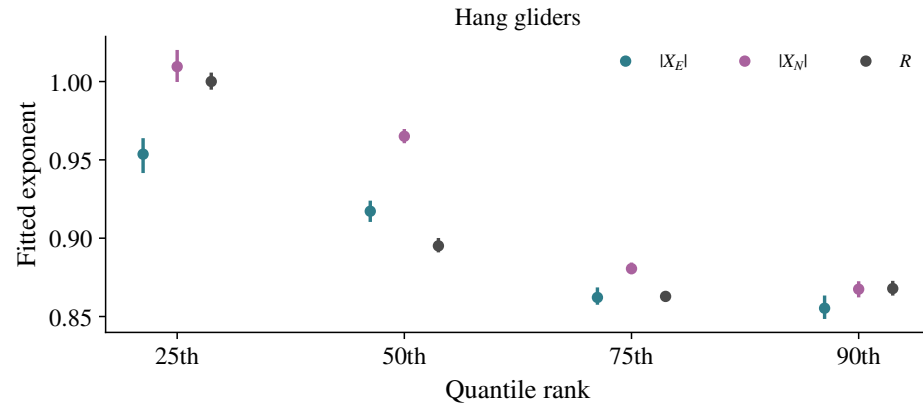
**Discussion.** Which differences persist under a change of fitting interval?

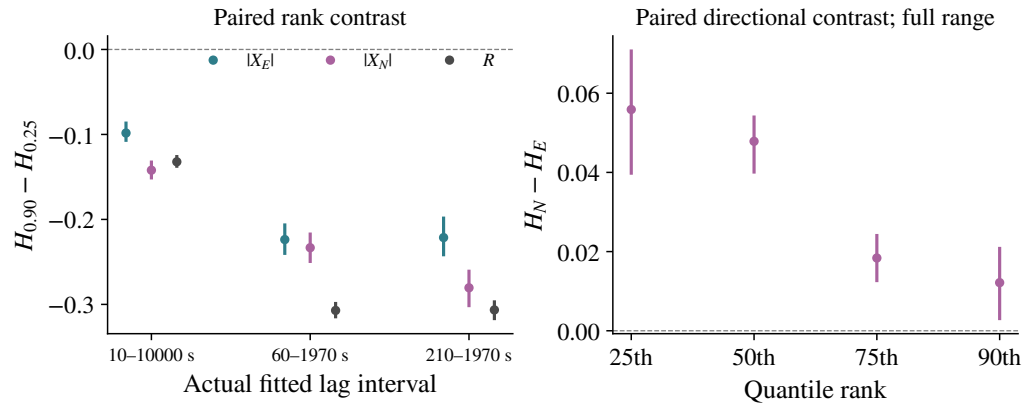
The full-range north-minus-east contrast is positive with a 95-percent paired interval above zero at every one of the four ranks, in both disciplines. The rank contrast is  $H_{90}$  minus  $H_{25}$  for east, north and radius over each saved fit range. The directional contrast is north minus east at each rank over the full range. Both retain pairing inside the flight bootstrap. The zero line marks the stated equality hypothesis, without a multiple-comparison calibration.

## Hang gliders: component and radial exponent estimates

**Discussion.** Which differences need a paired contrast instead of comparing overlapping intervals?

Markers are full-range fitted exponents; vertical segments are the exact saved 95-percent percentile endpoints from 400 whole-flight bootstrap draws. They are not simultaneous intervals and do not include sensor, site-day or fitting-range uncertainty.



**Hang gliders: paired rank and directional contrasts**

**Discussion.** Which differences persist under a change of fitting interval?

The full-range north-minus-east contrast is positive with a 95-percent paired interval above zero at every one of the four ranks, in both disciplines. The rank contrast is  $H_{90}$  minus  $H_{25}$  for east, north and radius over each saved fit range. The directional contrast is north minus east at each rank over the full range. Both retain pairing inside the flight bootstrap. The zero line marks the stated equality hypothesis, without a multiple-comparison calibration.

Thesis §3.3.2

## Distribution collapse

The separate  $H_{j,p}$  describe each curve. For the marginal collapse, fit one  $H_{j,*}$  per quantity, with a separate intercept for each rank:

$$\min_{H, \{b_p\}} \sum_p \sum_{\ell \in \mathcal{L}} \left[ \log \frac{Q_{j,p}(\tau_\ell)}{1 \text{ m}} - b_p - H \log \frac{\tau_\ell}{1000 \text{ s}} \right]^2.$$

Equal rank and lag weights on the same grid give  $H_{j,*} = \frac{1}{4} \sum_p H_{j,p}$ . With  $a_j(\tau) = [\tau / (1000 \text{ s})]^{H_{j,*}}$ ,

$$u = Y_j / a_j(\tau), \quad p_u(u; \tau) = a_j(\tau) p_{Y_j}(a_j(\tau) u; \tau).$$

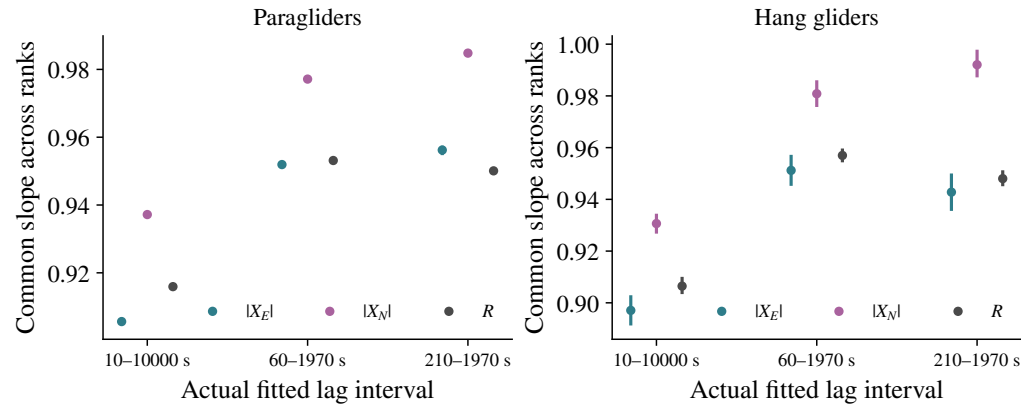
The horizontal scale and vertical density change together to preserve probability. The exponent is fixed before comparing the rescaled laws.

Thesis Sec. 3.3.2. Separate  $H_{j,p}$ , shared marginal  $H_{j,*}$  and the later joint-vector  $H_*$  serve different comparisons.

## The collapse exponent shares one slope across four quantile ranks

**Discussion.** Do the laws collapse under this candidate power, or do their shapes still change?

Averaging the four slopes is equivalent to the stated common-slope least-squares fit only because all ranks use the same lag grid with equal rank and lag weights and have separate intercepts. Sharing a fitted slope does not establish that the separate slopes are equal. The full-range common exponent is used for the power-rescaled marginal panels; narrower fitting windows provide sensitivity checks with the same observations. Dividing by each lag's own median is a further comparison that removes an arbitrary scale change, not only a power law. For the later signed two-dimensional maps, one scalar exponent averages all twelve separate slopes across quantities and ranks; it is not a lag-specific whitening or rotation.

**A common fitted scale depends on the fitted interval**

**Discussion.** Which range has an independent physical justification?

The component-specific common slopes average the four ranks with their own intercepts. The plot uses saved fits and intervals only, not new fitting. Agreement of a common summary does not establish equal individual rank slopes.

$$\hat{f}_j = \frac{p_j}{b_{j+1} - b_j}.$$

A very narrow histogram bin can have a tall density and still contain little probability. Across the displayed marginal laws, the mass below 1 mm, including zero and the straddling bin, is below 0.003%.

That numerical scale is not a claim of millimetre GNSS resolution. The following plots retain positive support; empirical-CDF comparisons also retain zero mass.

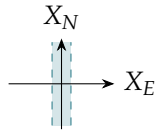
## A large density near zero need not carry much probability

**Discussion.** Do the central ranks and the probability-bearing tails tell the same scaling story?

The stated bound comes from the complete saved histogram masses, not from reading a logarithmic density height. Tiny numerical displacements can reflect recorded coordinates and interpolation. The median-normalized displays show the saved first-to-ninety-ninth percentiles; they do not establish tail collapse beyond those ranks.

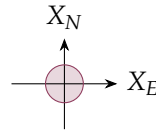
For a regular joint density, polar coordinates give

$$p_R(r; \tau) = r \int_0^{2\pi} p_X(r \cos \theta, r \sin \theta; \tau) d\theta.$$



A strip gives a component plateau:

$$p_{|X_E|}(x) \longrightarrow 2p_E(0) > 0.$$



A disk gives a radial rise:

$$p_R(r) \sim 2\pi p_X(0,0)r.$$

A rising radial density is compatible with exact scaling.

Finite, continuous, nonzero densities at the origin; no isotropy assumption. This explains the geometry, not a fitted small-radius law.

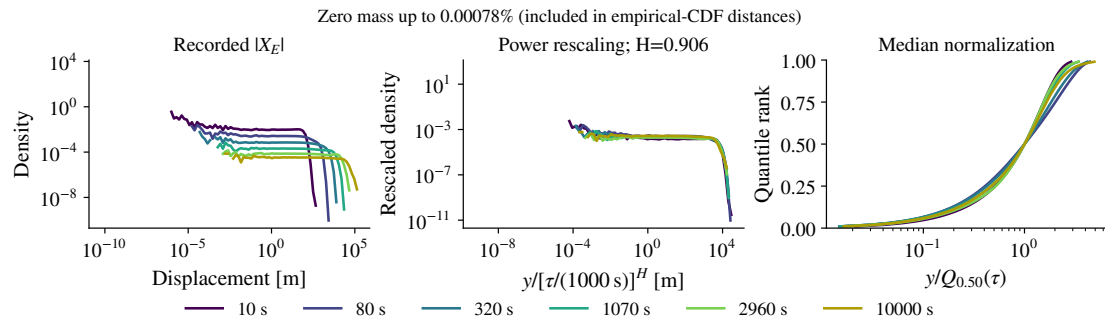
## A radial density can scale without a plateau near zero

The component density integrates probability over a strip; radial density integrates it around an annulus whose area includes the factor  $r$ . The corresponding cumulative probabilities near zero grow linearly and quadratically. Under a common spatial scale  $a$ , the factor  $r=a$  times the rescaled radius converts the joint density factor  $a$  to the minus two into the radial factor  $a$  to the minus one. This is consistent with the same one-dimensional scaling prefactor for the component and radial densities. As an exact example, independent centred Gaussian coordinates with common standard deviation  $\sigma$  give a half-normal absolute component with a plateau and a Rayleigh radius rising from zero; both collapse exactly when  $\sigma$  scales by  $a$ . The real flights need not be Gaussian, and their tiny-radius histograms need not reach these smooth-density limits.

**Paragliders: marginal collapse for east**

**Discussion.** Does power or median normalization leave a systematic shape change?

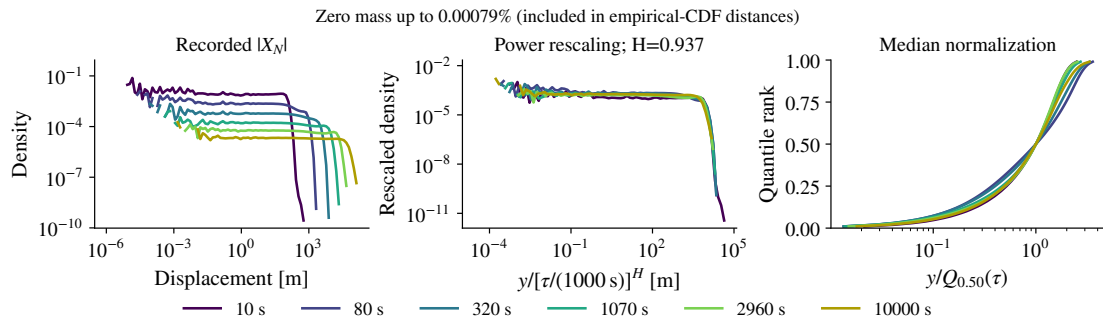
The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



**Paragliders: marginal collapse for north**

**Discussion.** Does power or median normalization leave a systematic shape change?

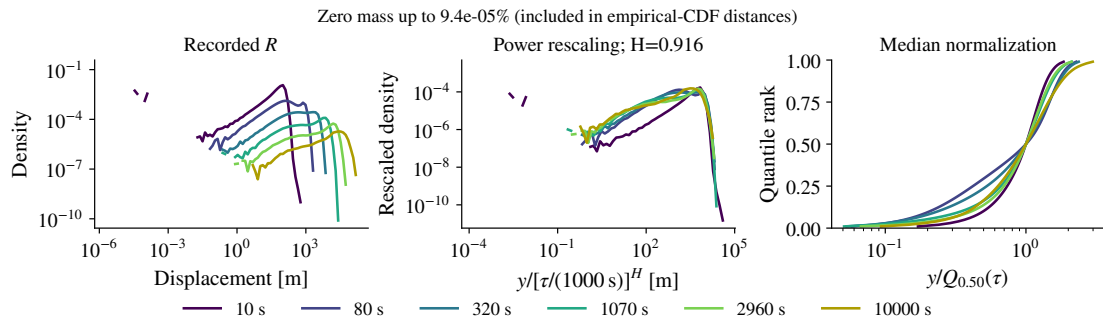
The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



**Paragliders: marginal collapse for radius**

**Discussion.** Does power or median normalization leave a systematic shape change?

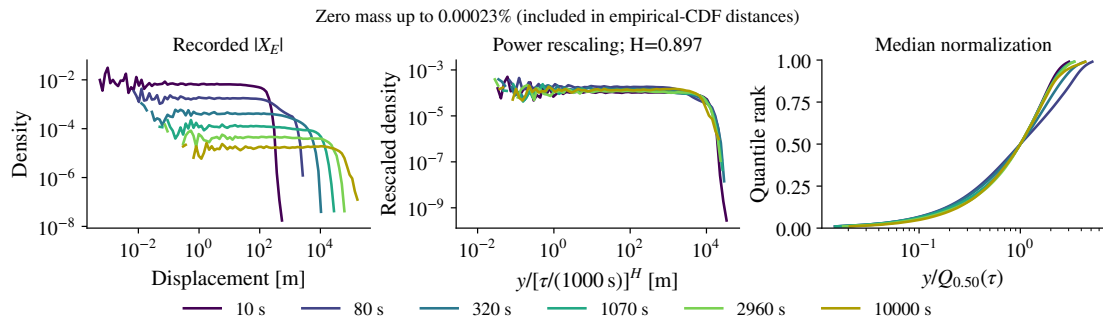
The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



**Hang gliders: marginal collapse for east**

**Discussion.** Does power or median normalization leave a systematic shape change?

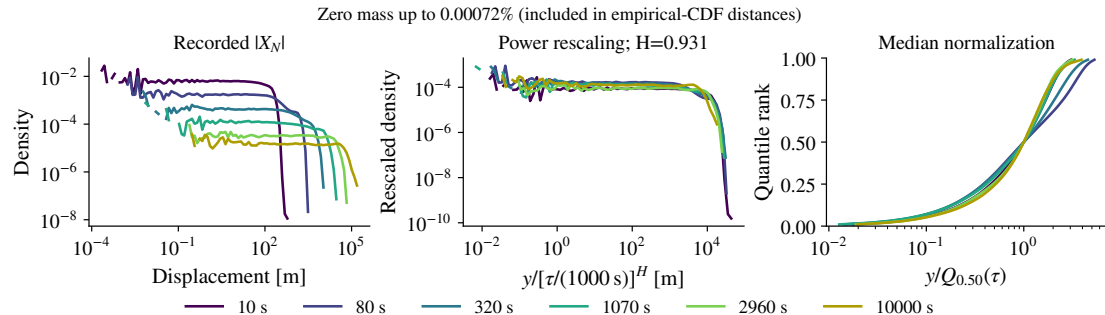
The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



## Hang gliders: marginal collapse for north

**Discussion.** Does power or median normalization leave a systematic shape change?

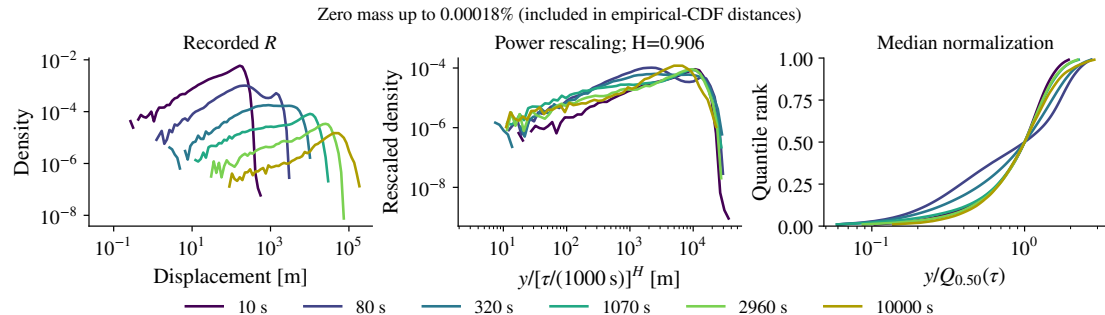
The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



## Hang gliders: marginal collapse for radius

**Discussion.** Does power or median normalization leave a systematic shape change?

The panels show recorded histogram densities, the saved power rescaling, and recorded first-to-ninety-ninth quantile ranks after median normalization. Positive values are displayed on logarithmic axes; any zero mass is reported separately and retained in the exact empirical-CDF distances. Median-normalized ranks do not establish collapse outside that rank range. The same selected lag colours are retained throughout.



Power rescaling visibly brings the absolute east and north densities together. Radial curves also group more closely, but differences remain around the peak and on its rising side.

| Maximum ECDF separation | $ X_E $ | $ X_N $ | $R$   |
|-------------------------|---------|---------|-------|
| Paragliders             | 0.177   | 0.203   | 0.273 |
| Hang gliders            | 0.185   | 0.269   | 0.331 |

A value of 0.177 means a 17.7-percentage-point cumulative probability difference for some pair of lags after power rescaling. Log density axes can conceal it.

**Conclusion:** [approximate scaling of absolute-component one-lag laws](#). These plots do not establish signed-component or multi-time process self-similarity.

Thesis Figs. 3.10–3.11 and Table 3.5. Fixed flights, weights and origins; descriptive distances without a calibrated rejection threshold.

## East and north show approximate collapse, with residual differences

**Discussion.** Which lag and probability ranges support a useful approximation, and what discrepancy should the model tolerate?

The visible component collapse should be acknowledged explicitly, while its quality is quantified rather than described as exact. The table contains the saved maxima over all pairs of the six displayed lags. Each quantity uses its own fitted common exponent; these are not a demonstration of one common vector exponent. The radial density's lack of a plateau is not evidence against scaling. Its separation between lags is the relevant discrepancy. Taking absolute values discards sign information, and separate one-lag laws leave temporal joint laws untested. Median normalization also leaves discrepancies: the radial maxima become 0.194 for paragliders and 0.229 for hang gliders. The distances are not independent-sample KS p-values and do not quantify statistical significance.

Thesis §3.3.3

## Signed joint distributions

$$\mathbf{Z}_\tau = \frac{(X_E, X_N)}{s_0 \left( \frac{\tau}{1000\text{s}} \right)^{H_*}}.$$

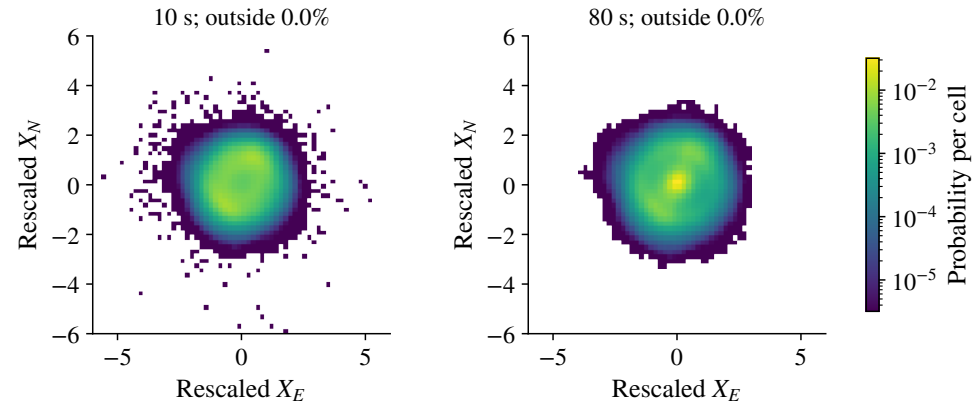
$H_*$  is the common fit across all component/rank slopes. The radial reference median near 1000 s is adjusted to 1000 s by the same power.

There is no lag-specific centring, rotation or whitening. The six lags share the same cells and equal total weight per flight.

## The joint comparison keeps geographic axes and one scalar scale

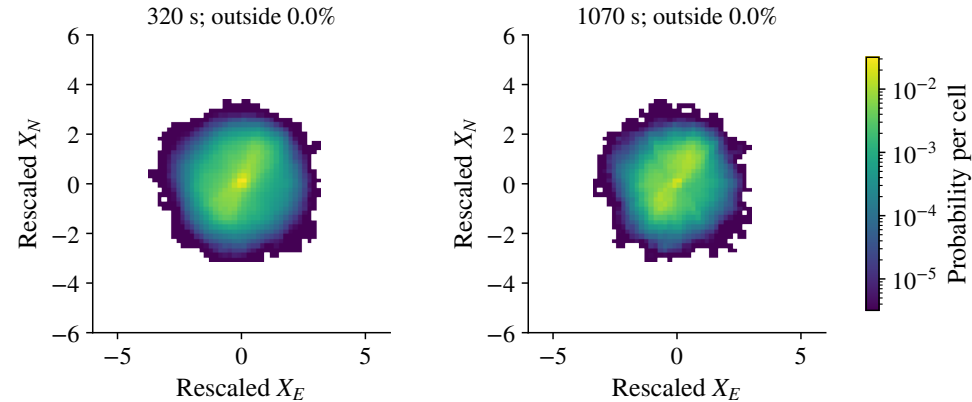
**Discussion.** Which directional and dependence changes remain visible when the normalization cannot remove them separately?

The common geographic axes are deliberate. Independent rotations or covariance normalization at each lag could erase part of the effect being studied. This is a test of one stated scale family of ground displacements, not a recovery of air-relative motion.

**Paragliders: signed joint laws (1/3)**

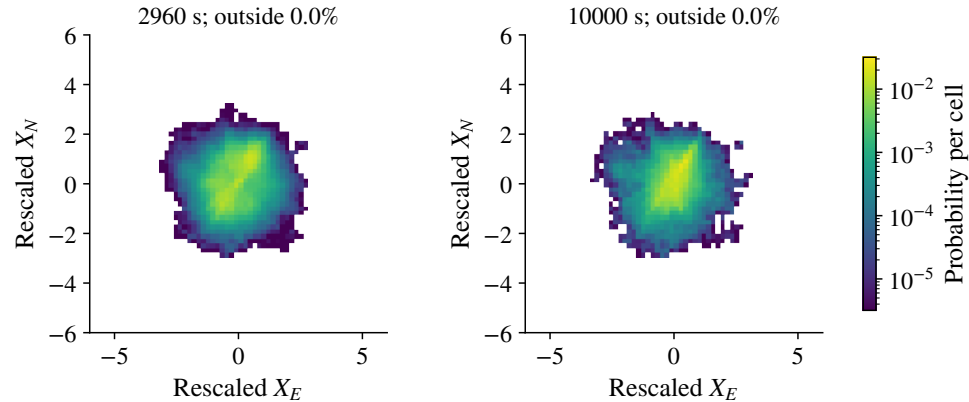
**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

**Paragliders: signed joint laws (2/3)**

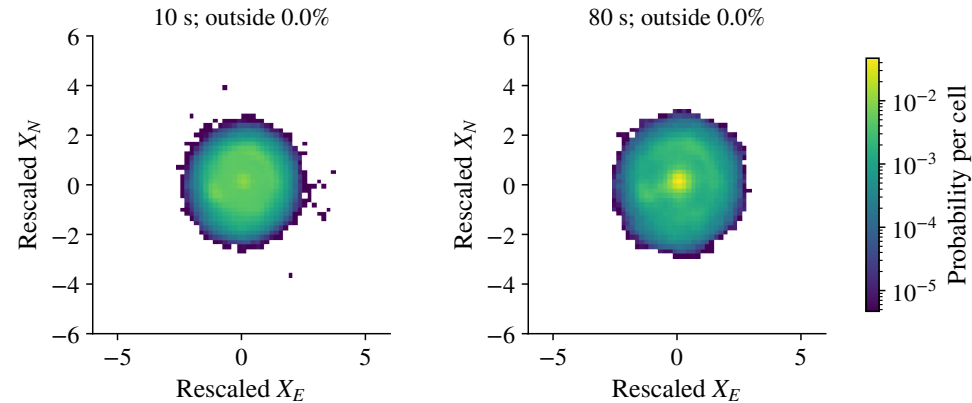
**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

**Paragliders: signed joint laws (3/3)**

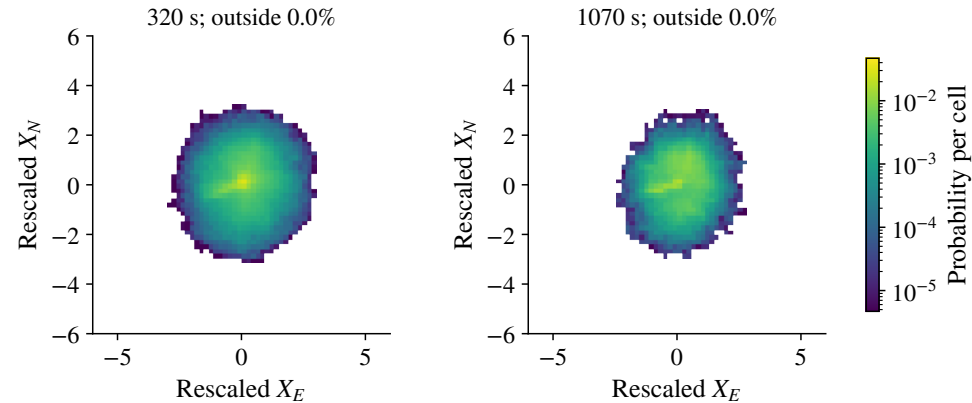
**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

**Hang gliders: signed joint laws (1/3)**

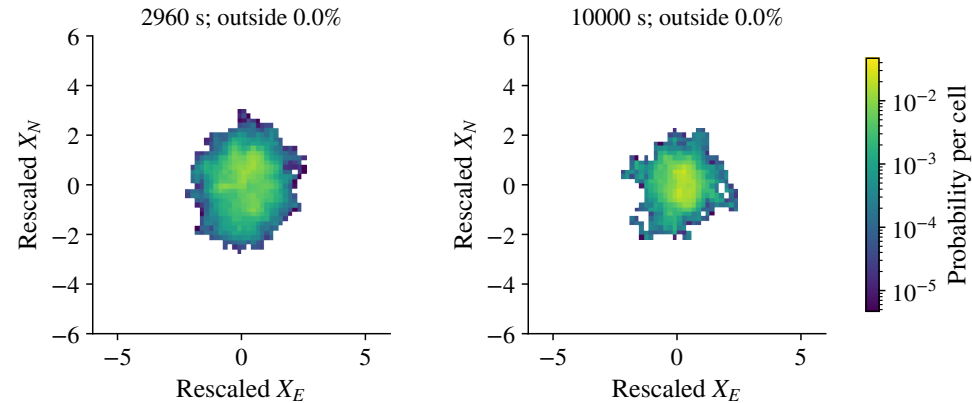
**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

**Hang gliders: signed joint laws (2/3)**

**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

**Hang gliders: signed joint laws (3/3)**

**Discussion.** Which changes concern orientation, central concentration or mass outside the displayed square?

Each discipline uses one common colour scale across all six lags, one geographic frame and one scalar rescaling. Colours are probability per original cell. Outside mass is retained in overflow cells for the numerical comparison. Equal plot area does not imply equal precision across disciplines.

|                                 | Paragliders | Hang gliders |
|---------------------------------|-------------|--------------|
| Common scalar exponent $H_*$    | 0.9196      | 0.9114       |
| Binned TV: 10 s versus 10,000 s | 0.370       | 0.403        |
| Largest TV among the six lags   | 0.370       | 0.441        |

Later rescaled laws concentrate more centrally, while their means also change. Overflow mass is zero to numerical precision in the displayed square.

Mean motion, centred spread, orientation and dependence can all contribute. The plot alone does not separate their physical causes.

## One scalar rescaling leaves changes in the signed joint law

**Discussion.** Would separating mean displacement from centred shape clarify which assumption fails?

The comparison uses fixed flights, common origins, equal flight weights, signed geographic components and the same bins. These distances describe the recorded finite distributions; they have no calibrated rejection threshold. The first-to-last pair is not the maximum pair for hang gliders.

$$D_{\text{joint}}(\tau, \tau') = \frac{1}{2} \sum_{ij} |p_{\tau,ij} - p_{\tau',ij}|.$$

The grid has 64 interior bins per axis and overflow cells. The distance resolves changes at that bin scale; it is not a calibrated rejection test.

Even agreement of every one-lag vector distribution would leave the joint law at several times, and hence full process self-similarity, untested.

## One-lag vector laws leave multi-time questions open

**Discussion.** Which multi-time or phase-conditioned comparison would most help choose between candidate processes?

Finite empirical distributions can differ under a common population law. Calibration must respect paired flights and dependent origins. Energy distance is an alternative multivariate diagnostic under finite first moments, but choosing another distance does not automatically supply the right uncertainty model.

Thesis §3.3.4

## Testing approximate scaling beyond one-lag magnitudes

$$\mathbf{X}_1 = \mathbf{r}(t + \tau) - \mathbf{r}(t), \quad \mathbf{X}_2 = \mathbf{r}(t + 2\tau) - \mathbf{r}(t + \tau),$$

$$\mathbf{Z}_\tau = (X_{1,E}, X_{1,N}, X_{2,E}, X_{2,N}), \quad \tilde{\mathbf{Z}}_\tau = \frac{\mathbf{Z}_\tau}{s_0(\tau/\tau_0)^{H_*}}.$$

An  $H$ -self-similar process with stationary increments predicts the same rescaled law at every lag.

$(\mathbf{X}_1, \mathbf{X}_2)$  also determines positions relative to the origin at  $\tau$  and  $2\tau$ :  $(\mathbf{X}_1, \mathbf{X}_1 + \mathbf{X}_2)$ .

Signs and the association between consecutive intervals are retained.

## Two consecutive displacements test temporal scaling

**Discussion.** Does agreement persist when a return or continuation is measured explicitly?

This is a new measurement on the verified cleaning 2.3.0 coordinates. The observable is four-dimensional. The transformation to two anchored positions is invertible. The pooled random origins do not themselves establish stationary increments; full process self-similarity would require all finite collections of times.

| Lag grid (s)             | Support through | Eligible para / hang |
|--------------------------|-----------------|----------------------|
| 10, 100, 1000, 10000     | 20000 s         | 14360 / 563          |
| 100, 200, 400, 800, 1600 | 3200 s          | 149831 / 5914        |
| 1000, 2000, 4000, 8000   | 16000 s         | 26977 / 1240         |

Every supported 10-s origin in every eligible segment; identical origins at all lags within a grid. Equal total weight per flight.

A fixed calendar-date split assigns whole days to training or validation. On training days, fit the four quantile ranks of  $|E_1|$ ,  $|N_1|$  and  $R_1$ ; average their twelve log–log slopes to obtain one  $H_*$ . Use the training radial median at the smallest lag as  $s_0$ .

The exponent and length scale are then fixed on validation days. No per-lag centring, rotation or whitening.

## Keep common support and separate training from validation

**Discussion.** How much do the different duration-supported populations affect the comparison between grids?

The protocol fixes the three grids before this measurement, informed by earlier descriptive results; endpoints are not estimated crossover times. All eligible flights with valid dates enter either training or validation. Three intermediate-grid paraglider flights have invalid catalogue dates and are excluded. Intermediate validation uses 75561 paraglider flights on 2997 dates and 2980 hang-glider flights on 938 dates. Training quantiles use 8192 logarithmic bins, with a 0.183 percent quantile resolution bound; validation CDF probabilities are directly counted. Training-fit uncertainty is conditioned on, because validation evaluates that fixed candidate on separate dates.

Four coordinate axes and all twelve normalised pairwise sums and differences give 16 fixed signed projections  $u$ . Evaluate their CDFs at

$$z \in \{0, \pm 0.1, \pm 0.25, \pm 0.5, \pm 1, \pm 2, \pm 4, \pm 8\}.$$

$$\hat{D} = \max_{\tau < \tau', u, z} |\hat{F}_{\tau, u}(z) - \hat{F}_{\tau', u}(z)|.$$

Resample whole validation dates 999 times, keeping the same multiplicity for every flight, origin, lag and projection within a day.

The 95th percentile  $c_{.95}$  of the maximum centred bootstrap error gives the upper bound  $U = \min(1, \hat{D} + c_{.95})$ .

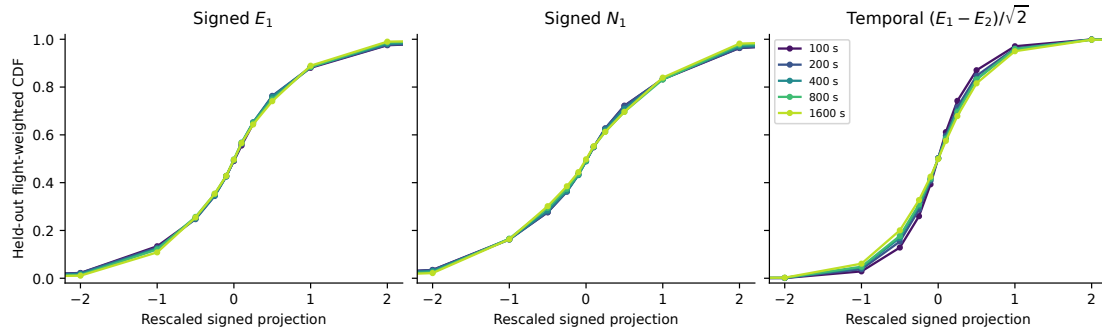
Require  $U$  below a declared tolerance: 0.05, 0.10 or 0.20.

Simultaneous within one discipline and grid; conditional on the training fit. Finite projections and thresholds, not an omnibus 4D test.

## Measure probability discrepancies and their uncertainty

**Discussion.** Which probability tolerance is acceptable for the model we want to build?

A discrepancy of 0.05 means five percentage points of cumulative probability, not five percent relative displacement error. The spatial group comprises E1, N1 and their sum/difference. The temporal group contains the remaining projections involving interval two. These include two-interval dependence, but a finite projection set cannot determine arbitrary multivariate laws. Cluster resampling assumes independently resampleable dates; weather across adjacent dates and repeat pilots can remain dependent. Bounds are not simultaneous across all six discipline-grid analyses and are not exact finite-sample guarantees. Positive evidence for approximate agreement comes from an upper bound below a stated tolerance, not from failing to reject exact equality.



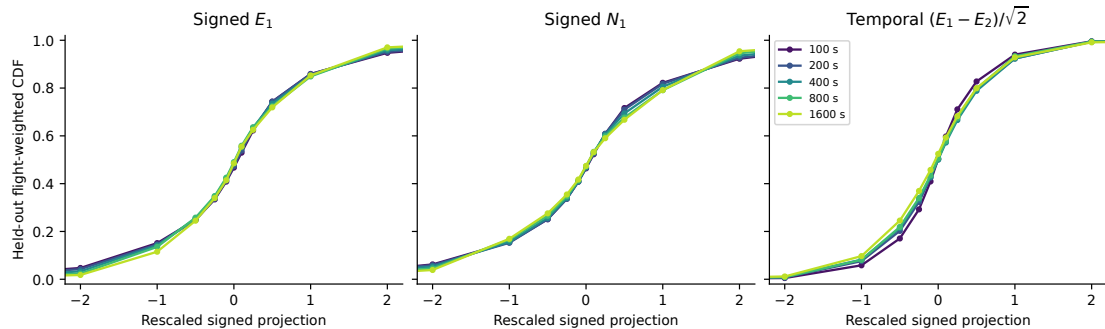
100–1600 s; training  $H_* = 0.915$ . Spatial-projection upper bound: 0.0313. Temporal upper bound: 0.0829.

The declared 0.05 spatial and 0.10 temporal tolerances are supported.

## Paragliders: intermediate scaling survives a temporal check

**Discussion.** Can phase mixtures explain why the temporal difference separates more than E or N?

The plotted CDFs use held-out dates, common origins and equal flight weights. All five lags are retained. Lines connect the tested thresholds; no claim is made about every intermediate threshold. The right panel is the predefined direction  $E_1$  minus  $E_2$  over square root two, which measures changes between consecutive eastward displacements. The reported temporal bound is the maximum over all twelve temporal projections, not just this displayed one. Exact equality is still inconsistent with the resolved nonzero discrepancies.



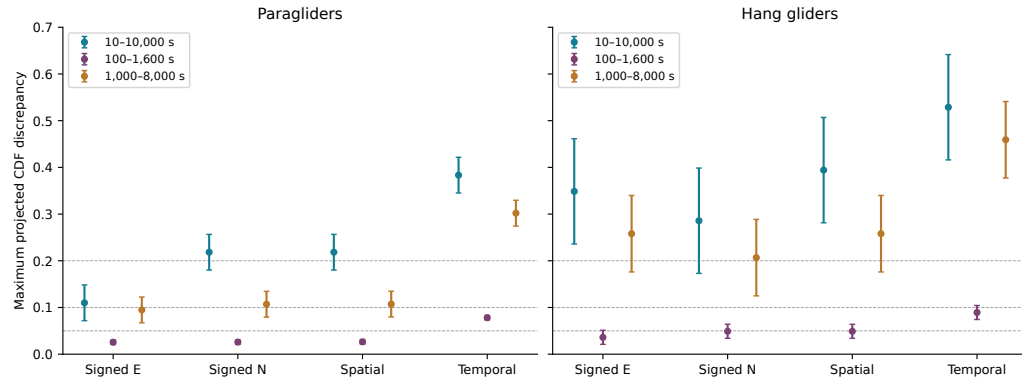
100–1600 s; training  $H_* = 0.899$ . Spatial upper bound: 0.0640. Temporal distance: 0.0893; upper bound: 0.1043.

Spatial agreement passes 0.10; the temporal result narrowly misses it.

## Hang gliders: similar collapse, less precise temporal agreement

**Discussion.** Does the remaining temporal discrepancy reflect flight behaviour, selection or limited independent days?

The temporal result passes the predeclared 0.20 criterion, but that is a substantially coarser statement. We do not round 0.1043 down to claim a 0.10 pass. The signed east and north upper bounds are 0.0511 and 0.0640. Fewer flights and dates contribute than for paragliders. Differences between disciplines do not by themselves identify a physical cause.



On the full and long grids, temporal lower bounds exceed 0.20 in both disciplines.

Points: maximum CDF differences. Bars: simultaneous 95% bootstrap bounds. Dashed lines: declared tolerances. Each grid has its own fixed cohort.

## The scaling approximation has a restricted lag range

**Discussion.** Would a common-cohort comparison distinguish scale changes from the duration selection between grids?

The full grid is 10, 100, 1000, 10000 seconds; the long grid is 1000, 2000, 4000, 8000 seconds. Temporal discrepancies are much larger than in the intermediate grid, and their lower bounds exceed twenty probability points. The populations differ between grids, though membership and origins are fixed within each grid. This prevents assigning the entire contrast causally to lag. The full and long comparisons do demonstrate a mismatch within each selected population.

### Evidence strengthened beyond visual collapse

Over 100–1600 s, one scalar exponent approximately preserves signed spatial and tested two-interval laws, most clearly for paragliders.

This supports an approximate self-similar description at the measured directions, thresholds, tolerances and population.

Exact invariance, arbitrary multi-time laws and increment stationarity remain unestablished. The fitted exponent is not an intrinsic Hurst exponent.

An intermediate interval between short manoeuvres and longer route constraints is a plausible interpretation; the mechanisms are not identified here.

### The supported claim is approximate intermediate-range scaling

**Discussion.** Is this approximation sufficient for a model, or should its temporal accuracy be the next target?

The positive conclusion is deliberately explicit: the work no longer stops at a visual resemblance of absolute components. It tests held-out signs, spatial projections and projections involving consecutive intervals, then quantifies an approximation error. It does not turn a finite grid into a proof of process self-similarity. The paraglider temporal upper bound is 8.29 probability points; the hang-glider bound is 10.43. These statements remain conditional on the cluster assumptions.

Thesis §3.4.1

## Flight weighting and a fixed-segment control

**Two groupings**

Alps, Pyrenees and Channel Coast use the defined geographic boxes.

Initial GNSS altitude  $h_0$  defines:

|                |                         |
|----------------|-------------------------|
| Plains         | $h_0 < 300$ m           |
| Hills          | $300 \leq h_0 < 800$ m  |
| Low mountains  | $800 \leq h_0 < 1500$ m |
| High mountains | $h_0 \geq 1500$ m       |

Native-grid TA-MSD; 155,085 para / 6,060 hang flights. Altitude groups include the full eligible archive.

**One flight-level estimator**

$$m_f(\tau) = \frac{\sum_s n_{fs} m_{fs}}{\sum_s n_{fs}},$$

$$M_g(\tau) = \frac{1}{N_g} \sum_{f \in g} m_f(\tau).$$

Pool supported origins inside each flight, then weight contributing flights equally.

Compare available segments with a fixed set supporting the largest lag.

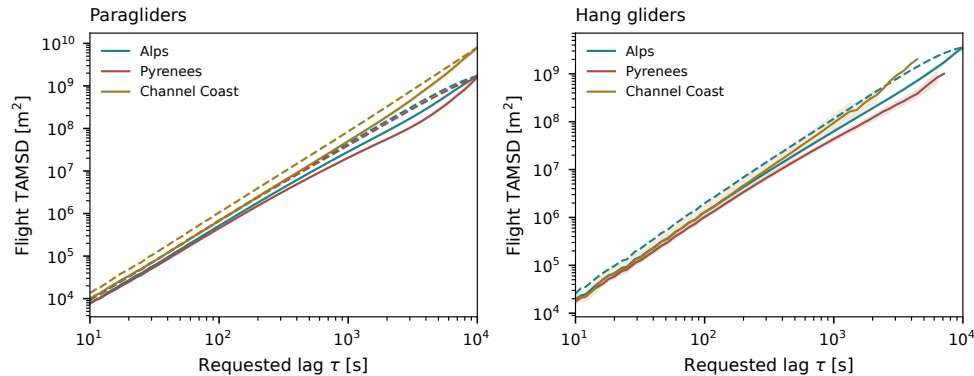
**Compare transport magnitude before its directional structure**

**Discussion.** How much of a group contrast concerns transport, and how much concerns which flights remain?

The altitude variable is the cleaned GNSS coordinate origin, with no pressure fallback. The chapter-two exploratory map used the raw first altitude with a fallback, so the two grouping sources must be distinguished. The familiar names do not measure relief, terrain encountered along a route or instantaneous flight altitude. Native intervals are at most ten seconds, each segment has at least eight fixes, and stored lag support ends at half the segment's grid length. Requested lags are rounded to native steps. No displacement crosses a gap. The fixed control retains identical segments and flights over all lags, but available time origins and their segment weights still vary.

Thesis §3.4.2

## Regional contrasts and changing population

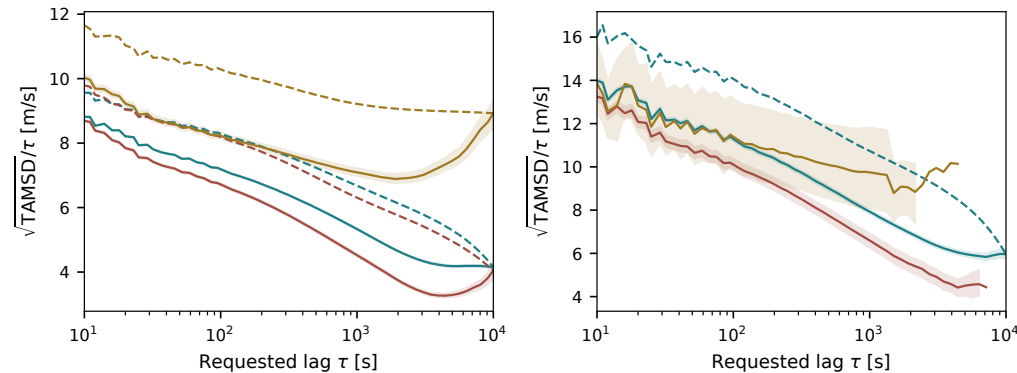


Solid: available segments. Dashed: fixed long segments. Shading: pointwise 95% site-day bootstrap intervals for solid curves.

## Regional TA-MSD: coastal flights travel farther at intermediate lags

**Discussion.** Does the coastal ordering persist after restricting all lags to the same long segments?

At the requested 1049-second lag, paraglider RMS displacements are 5.54 km in the Alps, 4.69 km in the Pyrenees and 7.42 km on the coast. The fixed-segment values are 6.97, 6.58 and 9.65 km. The coastal ordering remains, whereas the mountain contrast narrows. These are RMS endpoint displacements, not path lengths. Hang gliders have the same available-population ordering, but the coastal sample contains only 26 flights. Lags are the native TAMSD grid; 1049 seconds must not be labeled an exact 1000-second PCA lag.

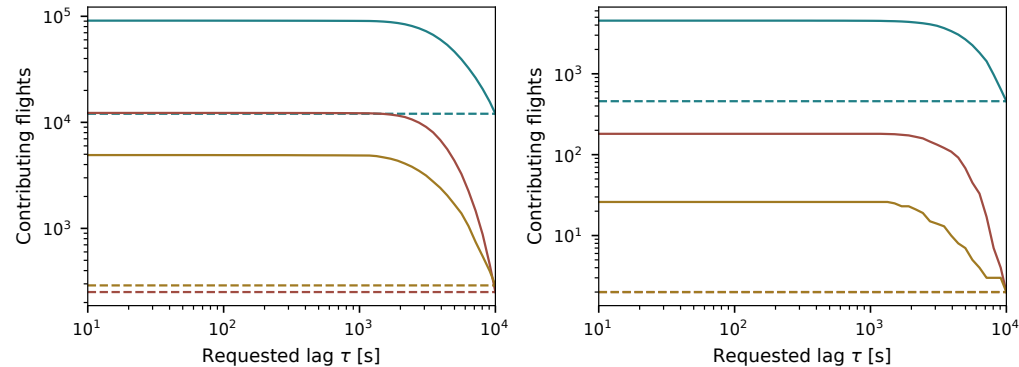


The ordinate is RMS interval-mean ground speed, up to native-grid rounding. A horizontal curve corresponds to quadratic TAMSD growth.

### Fixing the population changes the apparent long-lag growth

**Discussion.** Would the upward turn remain if short flights could contribute observed long-lag displacements?

The available coastal paraglider TAMSD has a descriptive long-range slope of 2.20; the fixed-segment curve has 1.98. The superquadratic appearance therefore does not persist in this fixed-population fitted comparison. It is not evidence for an intrinsic superballistic process. The Alpine and Pyrenean fixed curves continue to fall in this normalization. Their higher intermediate-lag amplitude also shows that long-flight selection changes the observed population before short flights lose support. These are descriptive interval fits, not Hurst estimates or calibrated significance tests.



Fixed paraglider flights: 12,053, 251, 290.

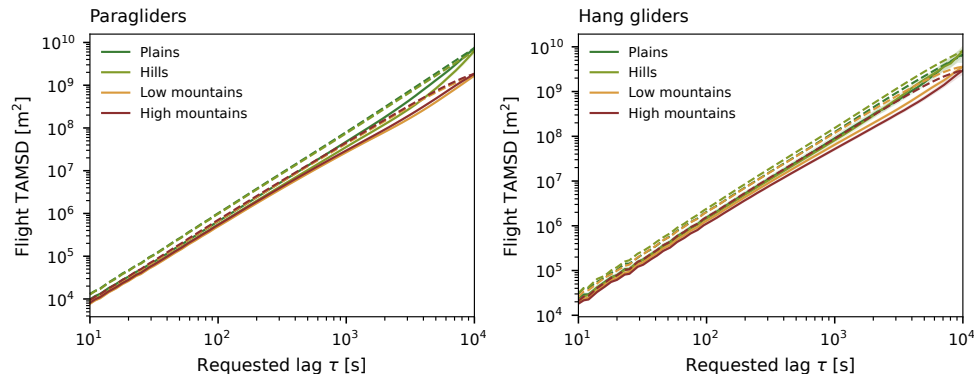
## Regional support limits what the fixed control can establish

**Discussion.** Which regional conclusions have enough long-flight support in both disciplines?

The solid curves show changing support and the dashed curves fixed membership. The fixed Alpine hang-glider control contains 458 flights, but the Pyrenean and coastal controls contain only two each. Those two-flight means are saved but omitted from displayed population curves. Eight flights is a display minimum, not a precision guarantee. Intervals require at least twenty site-day clusters; overlapping origins are never independent bootstrap units.

Thesis §3.4.3

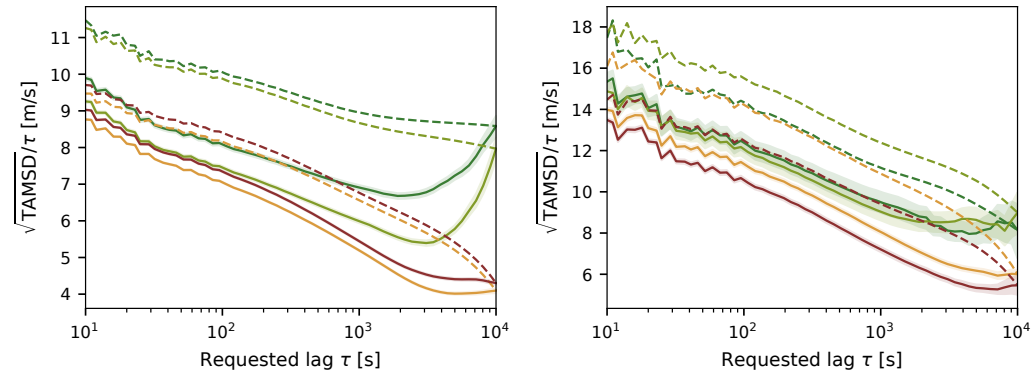
## **Initial-altitude bands and their regional composition**

**Initial-altitude groups combine transport and regional composition**

Bands use the cleaned GNSS origin: Plains, Hills, Low mountains, High mountains. All eligible regions enter these curves.

**Discussion.** Does an altitude-band contrast remain when region and contributing segments are held comparable?

At 1049 seconds the available paraglider RMS values are 7.23, 6.25, 5.40 and 5.66 km. The altitude ordering is not monotonic: the highest band exceeds the lower mountain band. For hang gliders the values are 9.91, 9.42, 8.38 and 7.49 km, but those groups differ in geographic and flight composition. Neither ordering is a within-pilot effect of altitude. The next panels expose the effect of fixing long segments and the lack of comparable regional-altitude populations.

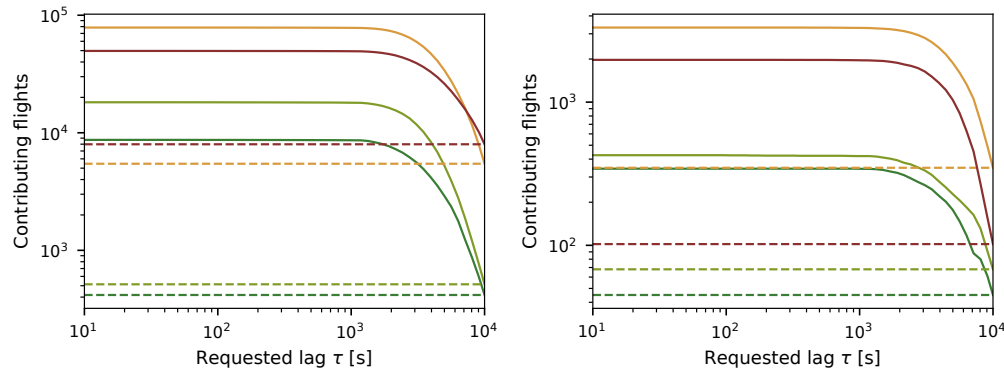


Solid: available segments; dashed: identical long segments at every lag. Shading is pointwise, not simultaneous.

## The late rise in Plains and Hills is sensitive to population selection

**Discussion.** Which parts of the ordering survive the fixed long-segment comparison?

Paraglider long-range slopes change from 2.19 to 1.97 in Plains and from 2.21 to 1.93 in Hills after fixing the contributing segments. Their RMS amplitudes remain above the mountain-labelled groups near the reference lag, but their separation becomes smaller. For hang gliders, the Plains/Hills ordering reverses in the long-segment population. These comparisons show selection sensitivity; they do not isolate an altitude or terrain mechanism.

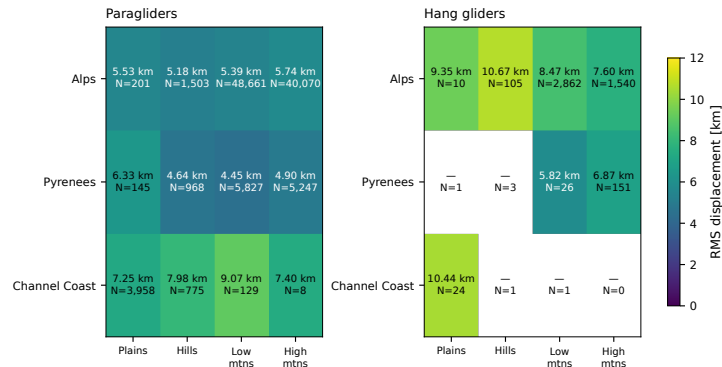


Counts concern supported flights. The fixed-segment control retains membership; its origins still depend on lag.

## The altitude-band comparison also needs its flight counts

**Discussion.** How much does the balance between altitude groups change as the lag increases?

Most eligible flights fall in the two mountain-labelled altitude bands, and long-flight retention differs greatly between bands. The paraglider fixed counts are 417, 514, 5452 and 7982; hang-glider counts are 45, 68, 348 and 102. The fixed control is conditional on long continuous segments, approximately twenty thousand seconds, with exact admission determined by the native-grid half-length rule. This explains small count differences from exact common-grid controls.



RMS displacement at the requested 1049-s lag; available segments. Descriptive cells, with no calibrated intervals.

## Region and origin altitude are strongly associated in this sample

**Discussion.** Can a pooled altitude contrast be interpreted as terrain when the same strata barely overlap?

The coastal sample mainly occupies Plains, while the mountain regions mostly occupy the two higher bands. Within Alpine paragliders, altitude-band amplitudes are much closer than in the pooled altitude comparison. The few coastal records with high initial altitude do not prove high terrain; records can begin airborne. Values below eight flights are omitted, with counts retained. Later, the coastal V2 change is smaller despite larger TAMSD: that is compatible with more persistent net advancement, but the two diagnostics use different lag grids and support, so it is a qualitative comparison. Independent wind and route information are needed for attribution. We now examine directionality, which a scalar TAMSD cannot identify.

Thesis §3.5

## Directional structure, terrain and equipment

| Diagnostic        | Clock                 | Weighting                  | Centring      |
|-------------------|-----------------------|----------------------------|---------------|
| Launch-time ratio | time since launch $t$ | paired flights at each $t$ | raw moments   |
| Regional PCA      | increment lag $\tau$  | eligible origins pooled    | mean removed  |
| Fixed joint law   | increment lag $\tau$  | equal total flight weight  | mean retained |

These measurements differ in method as well as population. Compare these choices before attributing a difference to a physical mechanism.

Regional plots use different clocks, weights and centring

**Discussion.** Which apparent disagreement remains after comparing like populations and observables?

The PCA draws on the full eligible regional archive, not only the fixed long-flight cohort. Longer supported records supply more increment origins to that pooled covariance. The launch-time ratios use a different clock and a changing population. Make these distinctions before linking the three figures to wind or terrain.

Thesis §3.5.1

**East–north balance in the retained archive**

$$\rho_{EN} = 1, \quad \Sigma = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix}, \quad \lambda_1/\lambda_2 = 19.$$

This centred Gaussian has balanced coordinate moments but strongly unequal principal variances.

A pooled mixture of east-directed and north-directed routes can also balance the ensemble while each flight remains directional.

### Equal east and north moments do not establish isotropy

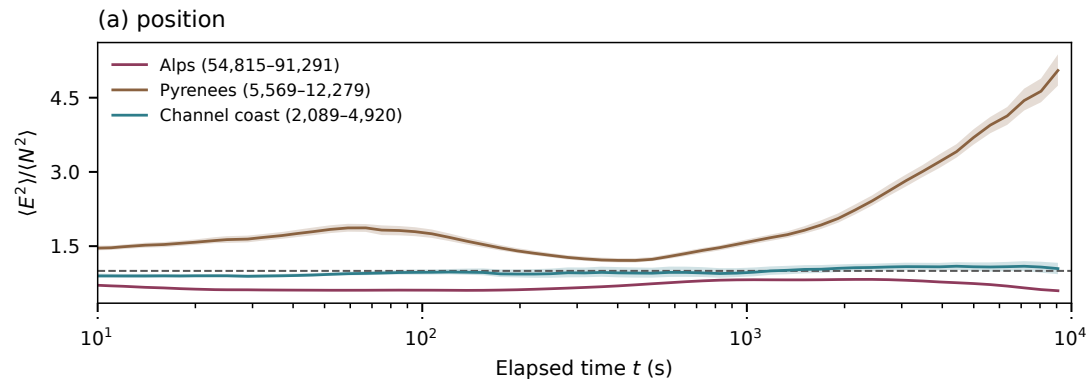
**Discussion.** Are we observing mean drift, centred directional variability, or a mixture of individually oriented routes?

Full isotropy requires rotation invariance of the law. An isotropic covariance alone would still be only a second-order condition. Raw moments additionally contain the outer product of the mean with itself. An intentional downwind route can be successful despite being directional, so this ratio is not a direct performance score.

**Regional position moments differ along east and north**

**Discussion.** How much imbalance reflects mean drift, route orientation or changing support?

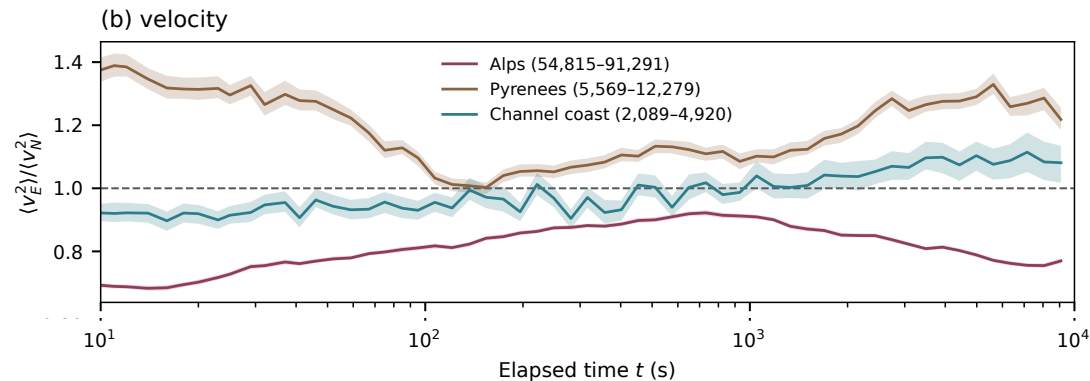
The clock is time since estimated launch, not an increment lag. Position ratios approach five in the Pyrenees. Raw second moments include both covariance and the squared mean. Splits prevent cross-gap stencils but recorded endpoint offsets can still affect these launch-referenced coordinates.



**Regional velocity moments show a related directional contrast**

**Discussion.** Which position and velocity contrasts persist under matched conditions?

The regional figure pools paired east/north observations at each elapsed time and uses site-day pointwise bands. Terrain, weather, routes and the disappearing short-flight population can all contribute. Coordinate-moment balance is weaker than full isotropy.



The ratios use paired east and north observations. A site–day bootstrap with 200 draws gives pointwise 10–90% ranges; each plotted time needs at least 30 flights and 10 clusters.

Missing site/date keys are treated as singletons. Resampling these groups does not match the equipment populations on weather, task or duration.

### Regional uncertainty bands do not match the groups

**Discussion.** Would a paired site–day comparison change the group difference and its uncertainty?

The bands are not 95-percent intervals, simultaneous bands or direct intervals for the difference between groups. Their overlap is not itself a calibrated significance test. The many displayed time points are correlated and must not be counted as independent confirmations of an ordering.

Thesis §3.5.2

## Regional principal axes

**A rotation preserves distance**

$$\Sigma = U\Lambda U^\top, \quad |U^\top \mathbf{X}| = |\mathbf{X}|.$$

PCA decorrelates components at a chosen lag. It cannot repair radial collapse or remove temporal dependence.

Whitening changes lengths and only fixes second moments at the fitted scale.

The following panels report regional PCA and its support. Anisotropy alone is compatible with a self-similar vector and a self-similar radius.

**Ground velocity has two unknown terms**

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

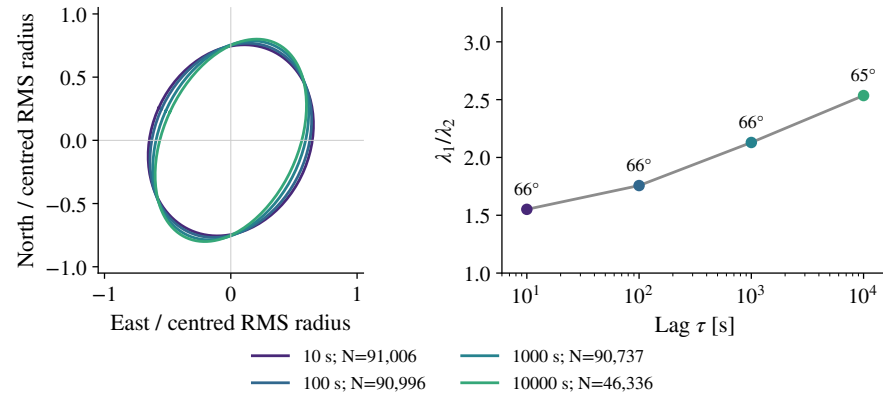
Terrain constrains headings; wind also changes position. Differentiation does not separate the two.

Different marginal exponents, or changing joint dependence, can prevent radial scaling.

**PCA describes preferred axes; wind needs independent information**

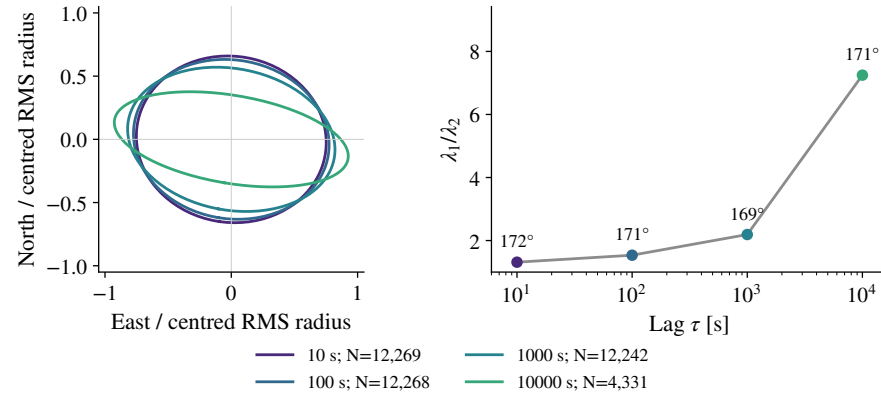
**Discussion.** Choose matched environments and an independent wind estimate before defining an “environment-free” model target.

Even independent Gaussian components can yield a changing radial exponent if their component exponents differ. If both component exponents agree, changing dependence can still prevent joint scaling. Conversely, a joint vector scaling law always implies radial scaling, whether the vector is isotropic or not. Rotating each flight randomly would balance aggregate directions without changing radial displacements; it cannot recover still-air motion. Independent wind information or carefully justified phase-based wind estimates should be discussed before choosing an environmental correction.

**Alps: centred principal axes across four lags**

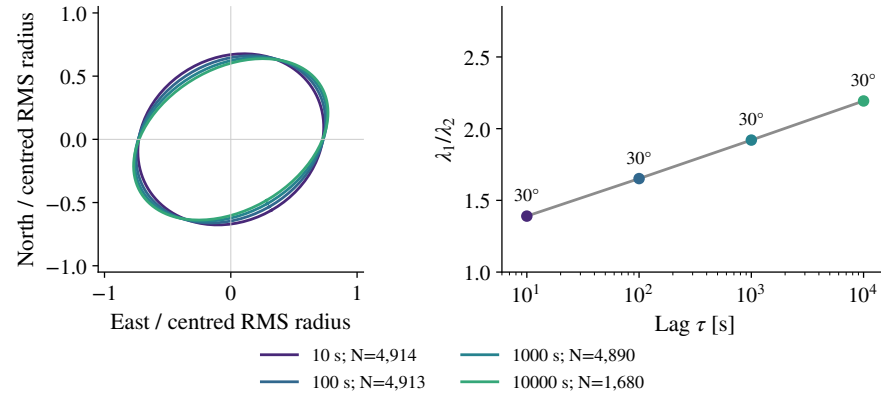
**Discussion.** Does the axis remain stable as lag and contributing population change?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

**Pyrenees: centred principal axes across four lags**

**Discussion.** How much of the stronger 10,000-s elongation persists when flight identities are held fixed?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

**Channel Coast: centred principal axes across four lags**

**Discussion.** Does the axis remain stable as lag and contributing population change?

The covariance pools eligible regional origins from the full archive. Ellipses are centred and normalized by the centred RMS radius; they hide absolute magnitude and drift. The axis is modulo 180 degrees and becomes poorly determined near equal eigenvalues. The lags are exactly 10, 100, 1000 and 10000 seconds, measured on the 10-second coordinate grid. Every supported flight contributes at each lag, and increments stay inside individual segments. The displayed counts shrink at long lags; covariance changes can therefore include population selection.

**NOAA ETOPO 2022**

Same Alpine and Pyrenean take-off boxes.

One-arc-minute elevation sampling; retain the footprint above 1000 m.

Project cell centres to regional east/north coordinates. The area-weighted spatial covariance gives the main axis.

NOAA NCEI, ETOPO 2022. Both elevation subsets, requests, code and hashes are archived. Angles are counterclockwise from east, modulo 180°.

**Definition sensitivity**

Thresholds: 500, 1000, 1500 m.

Alps: 42.1–47.9°.

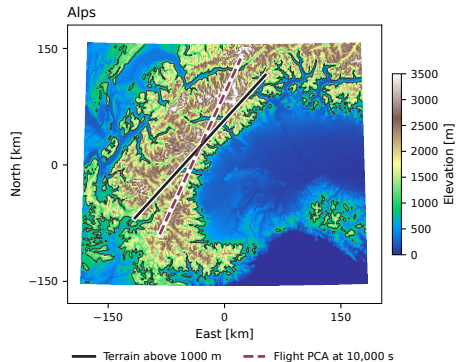
Pyrenees: 169.7–175.4°.

These spans are not confidence intervals; the axis describes an area, not a traced crest.

**A terrain direction measured independently of the flights**

**Discussion.** Would local axes along the actual tracks change the regional comparison?

Every fourth cell of the 15-arc-second grid gives one-arc-minute spacing. We select elevations at or above 1000 metres without fitting the threshold to the flight PCA. WGS84 tangent coordinates use the box centre and zero projection height. Weights are cosine latitude to approximate cell area; elevation above the threshold adds no weight. Threshold sensitivity is descriptive. The box boundaries, curved chains and plateaux affect the direction. Flight boxes identify take-offs, while increments can leave them and use flight-centred tangent frames. This is an indicative regional comparison.



Terrain:  $47.6^\circ$ .

Flight PCA:  $65.2^\circ$ .

Separation:  $17.6^\circ$ .

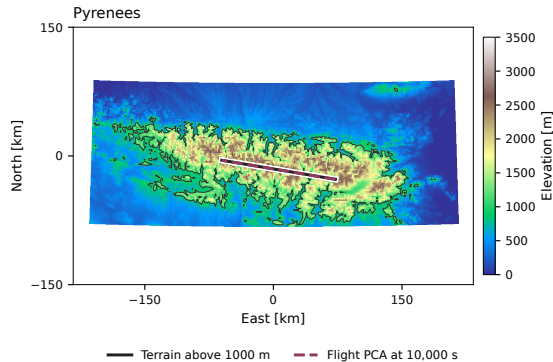
46,336 flights support 10,000 s.

Black: highland-footprint axis. Dashed colour: current flight PCA, shifted to the same centre. Line lengths have no physical meaning.

## Alps: shared sector, with an approximately 18-degree offset

**Discussion.** How much of the offset reflects Alpine curvature, box truncation or route selection?

Both orientations occupy the northeast-southwest sector, but the flight covariance is more northward. A simple claim that flights run parallel to one Alpine axis misses the approximately eighteen-degree difference. The terrain angle spans 42.1 to 47.9 degrees over the three thresholds, so this sensitivity does not eliminate the offset. Wind and long-flight selection remain possible contributors; the figure neither isolates a terrain effect nor tests pilot skill.



Terrain:  $170.5^\circ$ .

Flight PCA:  $170.5^\circ$ .

Separation: less than  $1^\circ$ .

4,331 flights support 10,000 s.

ETOPO footprint above 1000 m; threshold sensitivity:  $169.7\text{--}175.4^\circ$ . Decimal agreement does not imply this physical precision.

### Pyrenees: the long-lag spread follows the highland axis

**Discussion.** Does the alignment persist for fixed flights and for terrain sampled along their routes?

The near-parallel axes support a geographical association between displacement orientation and the range. They do not establish the causal contribution of terrain. Winds can align with the same range and pilots can select routes along it. The sensitivity span is not a sampling confidence interval. The comparison is made within the exact Pyrenean take-off box, but flight increments can extend beyond it.

**Same population as the 10,000-s PCA**

- 1,680 paraglider flights.
- 2,013 nonoverlapping windows, wholly inside retained segments.
- Mean cleaned GNSS altitude along each window; equal window weights.

Trapezoidal integration on the same 10-s segment grids. The target describes the supported long-lag windows; it is not take-off altitude or height above the terrain.

**Mean:** 1,006 m

Equal flight weights: 980 m.

Middle half of window means:  
854 m–1,160 m.

Approximate altitude above sea level;  
recorder datums are not harmonised.

**Channel Coast: the reference height follows the PCA windows**

**Discussion.** Is one representative height sufficient, or should wind be matched to the altitude and location of each flight window?

The mean is 1006.262 metres across 2013 exact PCA windows from 1680 flights, with the same equal-window weighting as the covariance. Each window altitude is a trapezoidal time average of cleaned GNSS  $z$  interpolated onto the existing 10-second grid. No gap is bridged. Equal weight per contributing flight gives 980.260 metres; the window-mean quartiles are 853.932 and 1159.549 metres. Short flights do not define this target because they cannot contribute at 10000 seconds. The raw GNSS vertical datum is not harmonised, so MSL is an approximation; these numbers do not imply metre-level physical accuracy. The quartile range describes heterogeneity, not uncertainty in the mean. Matching the target height does not yet match flight dates, instantaneous positions or local atmospheric conditions.

**2016–2025, hourly; nine cells in the coastal box; 1000/925/850/700 hPa.**

$$H = \Phi/g_0, \quad h = \frac{R_{\oplus}H}{R_{\oplus} - H}, \quad \theta_w = \text{atan2}(\langle v(h) \rangle, \langle u(h) \rangle).$$

Convert geopotential to each hour's level heights. Interpolate **E/N wind components** to 1,006 m, between above-ground bounds.

**Main comparison: all year, all hours vs. all year, 09–17 UTC.**

April–September is a separate seasonal check. Winds at 10/100 m AGL provide lower-level controls.

ECMWF/Copernicus ERA5 pressure levels, accessed through a pinned snapshot of Earthmover's public cloud-optimised ERA5 copy on AWS. Earthmover distributes this lossless copy; it is not a separate reanalysis. Complete hourly extract: 32.22 MB compressed, preserved in the repository and on SSD.

## ERA5 wind is interpolated to the measured flight altitude

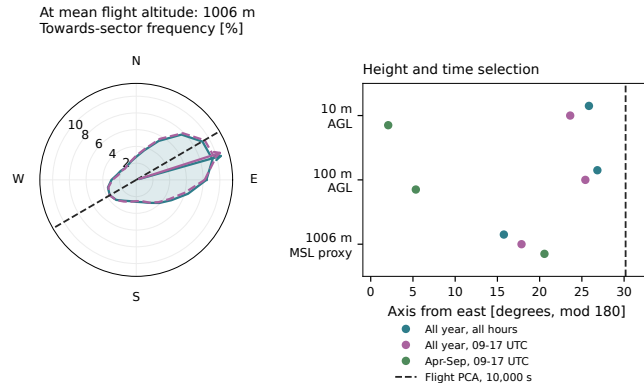
**Discussion.** How much of the surface-to-flight-height contrast reflects vertical shear, season and the flight conditions represented?

Use the same nine grid cells as the surface analysis: latitudes 48.75, 49.75, 50.75 degrees and longitudes minus 1.25, zero, 1.25 degrees. Geopotential height is Phi divided by 9.80665; geometric height uses spherical Earth radius 6371000 metres. A pressure level changes altitude from hour to hour. Linear interpolation acts on eastward and northward velocities, never direction angles. All 789048 cell-hour profiles bracket the main target above the model surface. Average components over time and with cosine-latitude cell weights, then reduce the towards-angle modulo 180 degrees for comparison with PCA. The annual all-hour selection has 87672 hours per cell; 09 through 17 UTC inclusive has 32877. The two warm-season selections have 43920 and 16470. UTC is fixed, without a daylight-saving correction. Source values, timestamps, coordinates, pressure levels, geopotential, surface geopotential and hashes are preserved losslessly for offline reuse.

**At flight height, both annual references remain offset from PCA**

**Discussion.** Why is the seasonal contrast larger near the surface?

At the mean flight altitude the all-year all-hour axis is 15.8 degrees and the 09–17 UTC axis is 17.9 degrees, against 30.2 degrees for flight PCA. The corresponding separations are 14.4 and 12.3 degrees. At 10/100 metres AGL the annual axes lie closer to PCA, but their warm-daytime references rotate further away. At flight altitude the warm-daytime axis is 20.6 degrees, giving a smaller 9.6-degree separation. The altitude-matched comparison therefore has a less close annual alignment and a smaller time-selection sensitivity than the surface comparison. These are descriptive contrasts, not formal tests of different angles. The rose uses weighted 15-degree sectors, excludes speeds below 0.5 m/s only from frequencies, and retains all hours in vector means. Teal is all-year all hours; plum is all-year daytime. Green appears only in the right-hand seasonal check.



Annual gaps at flight height: 14.4° for all hours, 12.3° for 09–17 UTC.

Angles counterclockwise from east. The rose shows where air moves towards at 1,006 m; the black line is the 30.2° centred PCA axis.

| At 1,006 m          | Wind axis | PCA gap | $\rho_w$ |
|---------------------|-----------|---------|----------|
| All year, all hours | 15.8°     | 14.4°   | 0.43     |
| All year, 09–17 UTC | 17.9°     | 12.3°   | 0.44     |
| Apr–Sep, all hours  | 16.0°     | 14.2°   | 0.35     |
| Apr–Sep, 09–17 UTC  | 20.6°     | 9.6°    | 0.37     |

At the quartile heights, annual axes span 14.3–17.2° (all hours) and 16.4–19.3° (daytime), on shared support.

$\rho_w = \|\langle \mathbf{w} \rangle\| / \langle \|\mathbf{w}\| \rangle$  measures cancellation. Height ranges are sensitivities, not confidence intervals. The main estimates retain all selected hours.

Height and time checks preserve a broad orientation resemblance

**Discussion.** Does the remaining offset reflect flight selection and route choices, or change when weather is matched to actual flights?

The annual all-hour and daytime resultant fractions are 0.43 and 0.44; warm-season values are 0.35 and 0.37. These do not give confidence in a direction or the fraction of observations near it. At the lower quartile height, 868 profiles at one cell have no above-ground pressure level below the target. For the height sensitivity only, exclude these 868 hours from all nine cells at every tested height, including the mean-height reference: 86804 common hours remain per cell. Apply the same seasonal and daytime masks afterward. On common support the annual quartile-height axes range from 14.3 to 17.2 degrees for all hours and from 16.4 to 19.3 degrees for daytime. Equal-flight mean altitude changes the matched reference by less than 0.3 degrees, and common-hour selection changes the main annual reference by less than 0.1 degrees. These checks retain the offset but do not calibrate receiver datums or establish significance.

| Region        | Conclusion supported by this comparison                                                                                                    |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Pyrenees      | Close terrain–flight alignment; a geographical association worth testing along tracks.                                                     |
| Alps          | Same broad sector, with an approximately 18-degree offset from the mean highland axis.                                                     |
| Channel Coast | Wind at mean flight altitude lies about 12–14 degrees from PCA in both annual time selections; seasonal checks retain a broad resemblance. |

A common constant wind shifts the mean displacement by  $\tau\mathbf{w}$ ; centring removes it. Variable winds and route responses can still shape covariance.

Independent regional references constrain an interpretation. They do not separate wind, terrain, route choice and long-flight selection.

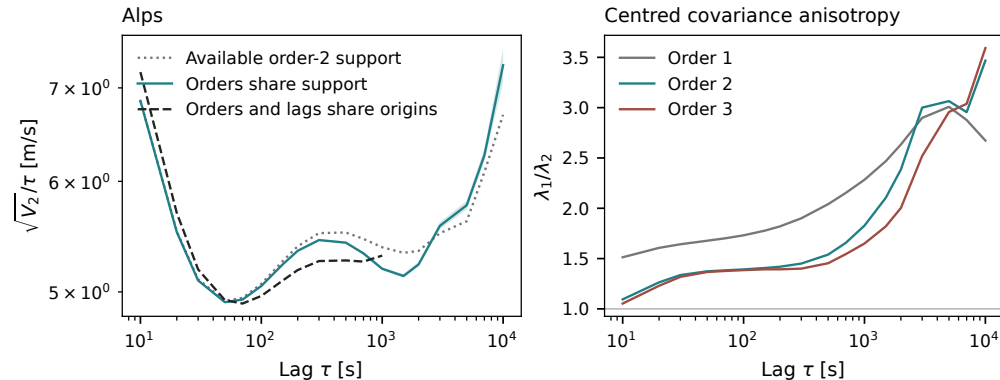
What the environmental comparisons establish

**Discussion.** Match flight times, heights and routes to wind and local terrain before attributing the ellipse or equipment contrasts to a cause.

The PCA concerns centred displacement spread, whereas the wind reference concerns a mean velocity. Their orientation resemblance alone cannot show that mean wind causes covariance anisotropy. Constant additive drift cancels when displacement is centred at fixed lag. Variable drift between flights, time-varying forcing and route decisions can alter covariance; their contributions need matched observations. The Pyrenean agreement is the strongest geographical alignment here. The coastal comparison at mean flight altitude retains a broadly similar sector across the four time selections, with offsets of about 10 to 14 degrees. Surface winds exhibit a larger seasonal change, so their annual near alignment should not be transferred to flight altitude. Equipment class is not measured pilot expertise, and this comparison cannot establish resistance to wind.

Thesis §3.5.2

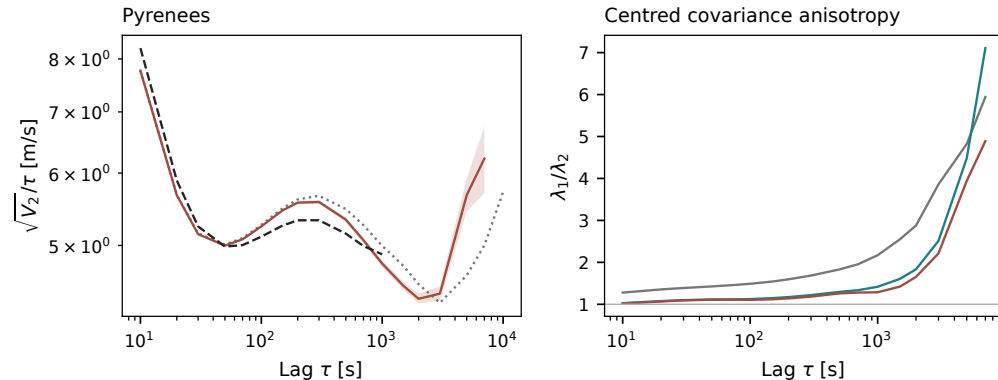
## Regional changes of velocity and direction

**Alps: velocity changes retain a directional structure**

Equal flight weights. At 1000 s, matched orders retain 89,181 flights. Shading: pointwise 95% site-day intervals.

**Discussion.** Which part of the lag dependence persists with common origins and after changing difference order?

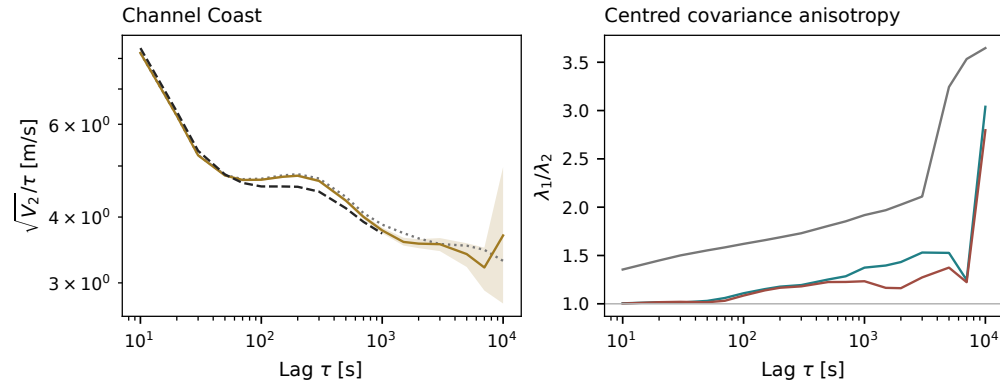
Left: available-order-two origins dotted, matched orders solid, and common origins through 1000 seconds dashed. Right: covariance anisotropy of orders one, two and three on identical origins. Unlike the preceding pooled-origin PCA, these curves give equal weight to flights. The full report also saves the pooled-origin sensitivity. Eighteen lags cover 10 to 10000 seconds. Matching all orders requires three times the lag; common origins use every ten-second origin supporting a 3000-second stencil. This changes the observations as well as flight membership.

**Pyrenees: the long-lag change coincides with falling support**

At 1000 s: 12,020 matched flights; at 10,000 s: only 3. Cells below eight flights are omitted from plots.

**Discussion.** How much of the late rise survives a comparison with fixed flights and origins?

The conventions are identical to the Alpine panel. At 1000 seconds the order-two eigenvalue ratio is 1.42, compared with 2.17 for ordinary increments. The sharp late changes need to be read together with the support figure. All numerical cells are retained in the report, including the final three-flight cell, but that cell is not plotted as a population result. The omitted 10000-second order-two ellipse is therefore intentional.

**Channel Coast: smaller changes at 1000 s, with residual anisotropy**

At 1000 s: 4,679 matched flights,  $D_2 = 3.77$  m/s and covariance ratio 1.37.

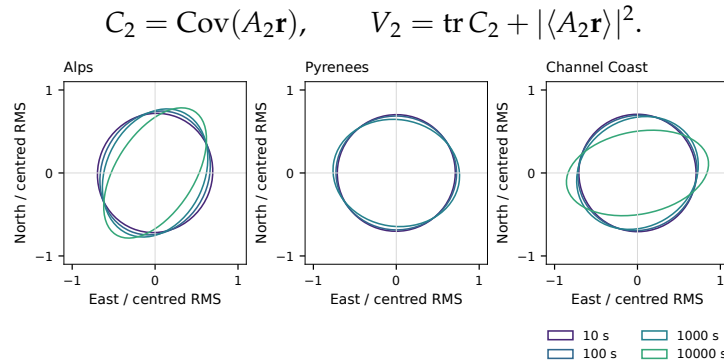
**Discussion.** Could route geometry and varying forcing explain the contrast that survives constant-velocity cancellation?

The coastal RMS change becomes lower than the mountain values over part of the lag range. It is higher than the Alpine value at ten seconds, so one amplitude ratio does not describe every scale. An added constant wind alone cannot explain a  $V_2$  difference between otherwise identical populations. Variable wind, its effect on decisions, route geometry and selection remain possible. These boxes classify take-off locations, not exposure along the whole flight.

## Second-difference ellipses retain orientation after drift cancellation

**Discussion.** Do the resolved axes align with independent terrain and wind information?

The four colours match the exact PCA display lags. Ellipses describe centred spread and omit both amplitude and mean. Near-circular ellipses, especially the coastal ten-second one, do not resolve a useful direction. At 1000 seconds the order-two axes are approximately 65 degrees in the Alps, 171 degrees in the Pyrenees and 31 degrees on the coast, counterclockwise from east. Axes are modulo 180 degrees and do not supply signed wind directions. The preceding independent comparisons provide regional references; they do not measure terrain or weather exposure along individual tracks, and the coastal mean-height wind reference retains a 10–14-degree offset across the tested time selections.

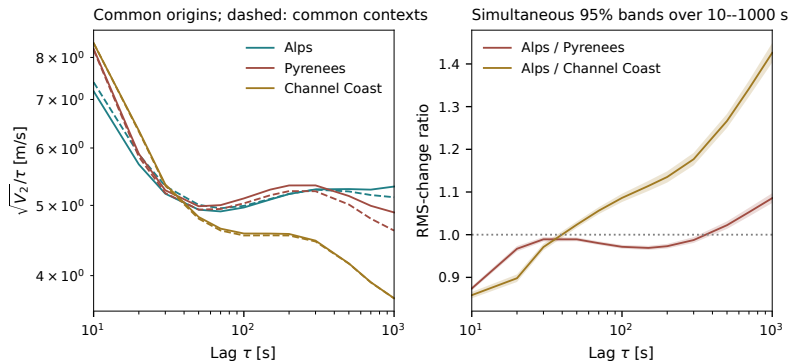


Matched-order origins, equal flight weights; each ellipse is normalised by its own centred RMS.

**The regional contrast persists in a common-context control**

**Discussion.** Which part survives matching actual weather, sites and pilot identity?

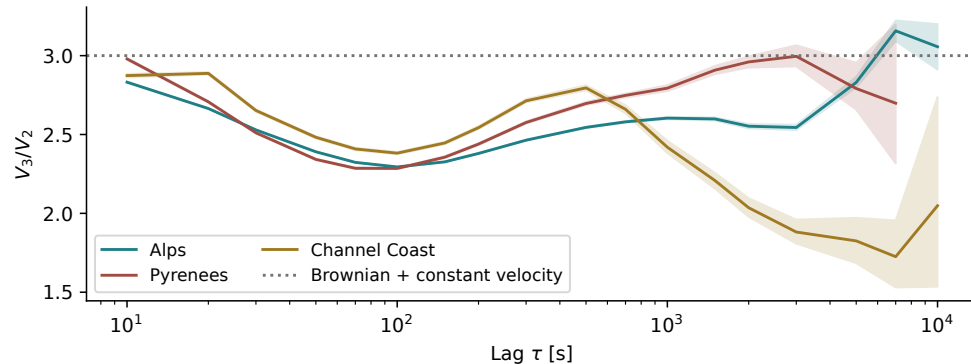
The context control standardises task, retained-duration band, season period and equipment, with at least ten flights per region per retained stratum. Target stratum mass is proportional to the smallest regional count. It retains 65537 Alpine, 9933 Pyrenean and 4366 coastal flights. The 1000-second RMS changes are 5.13, 4.62 and 3.72 m/s. These dashed point estimates have no calibrated intervals. The shaded ratio intervals instead apply to the unstandardised common-origin population and use 500 site-day bootstrap replicates jointly over the supported lag grid. The Alpine-to-coastal ratio there is 1.426 with a simultaneous band from 1.403 to 1.449 at 1000 seconds. No equivalence margin was specified, so a non-significant difference would not establish equality.



Common origins through 1000 s. Dashed: 23 shared context strata, descriptive. Shading: simultaneous 95% bands for the unstandardised ratios.

Thesis §3.5.2

## Third differences and the distribution of changes



Matched origins and flight weights; pointwise 95% site-day intervals. Dotted: Brownian motion plus constant velocity.

### The ratio between orders changes with the lag

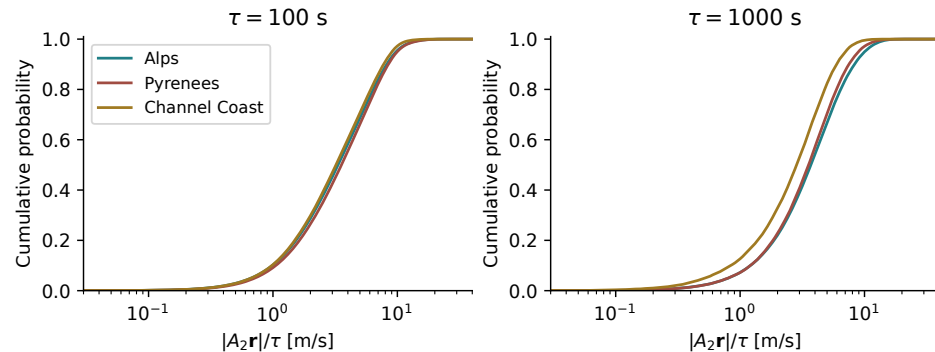
**Discussion.** Which scale interval could satisfy both the slope and amplitude predictions of one model?

$V_3/V_2$  is approximately 2.29 in both mountain regions at 100 seconds; at 1000 seconds it is 2.60 in the Alps, 2.79 in the Pyrenees and 2.42 on the coast. The lag dependence is itself diagnostic.  $V_3$  is not an independent sample and does not automatically isolate still-air dynamics. Its longer support and larger sensitivity to position noise must accompany any interpretation. No Hurst estimate is validated by these curves.

## The distribution distinguishes frequent small changes from large events

**Discussion.** Where on the trajectories do the largest contributions occur, and which verified flight phases contain them?

At 1000 seconds the upper tenth of changes contributes about 43–47 percent of the squared-change signal in the Alps, 40–44 percent in the Pyrenees and 38–43 percent on the coast. Those ranges bound the unresolved histogram boundary bin; they are not sampling confidence intervals. The result measures concentration of this observable. It does not label turbulence or a particular flight phase. Phase conditioning and independent wind attribution have not been performed in this extension.

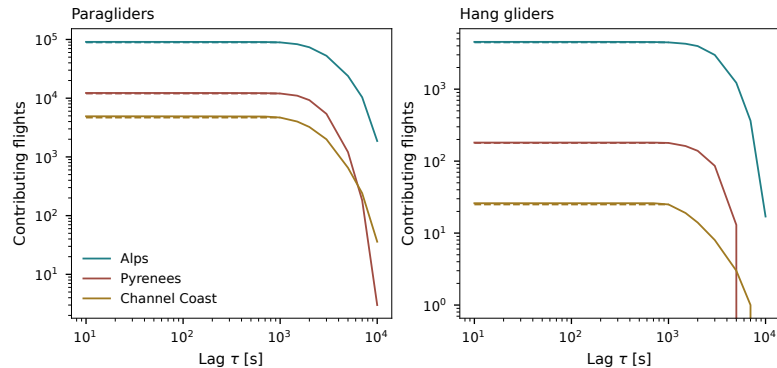


Equal-flight CDFs on matched-order support. Full report: all four display lags, zero and overflow mass, and quantile bin bounds.

**Order-three support limits the long-lag regional comparison**

**Discussion.** Which conclusions remain supported in both disciplines and across the same range of scales?

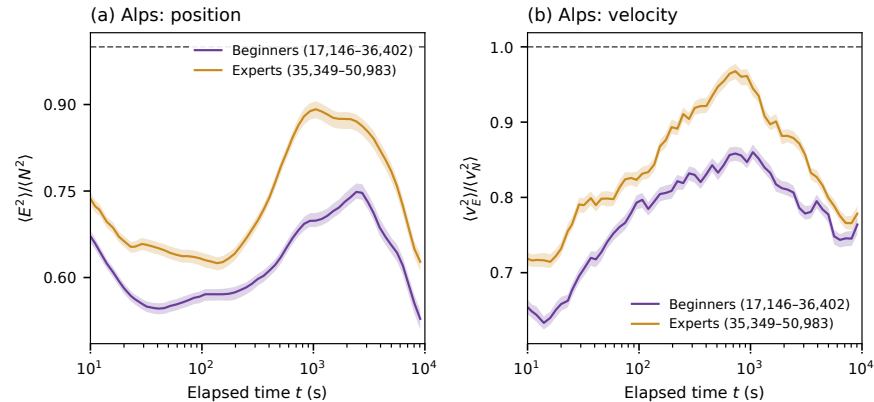
The full regional measurement includes 108189 paraglider flights and 4747 hang-glider flights. At 10000 seconds, matched paraglider counts are 1865, 3 and 36 for Alps, Pyrenees and Channel Coast; hang-glider counts are 17, zero and zero. Coastal hang gliders number only 26 even at short lags. The report retains their estimates and suppresses intervals below twenty contributing site-day groups. No common three-region equipment/context strata survive for hang gliders, so that standardisation is not claimed for this discipline.



Solid: orders share  $3\tau$  support. Dashed: origins common through 1000 s. All eligible regional flights contribute where supported.

Thesis §3.5.3

## Terrain, equipment and wind

**Alps: C/D/CCC has smaller coordinate-moment imbalance**

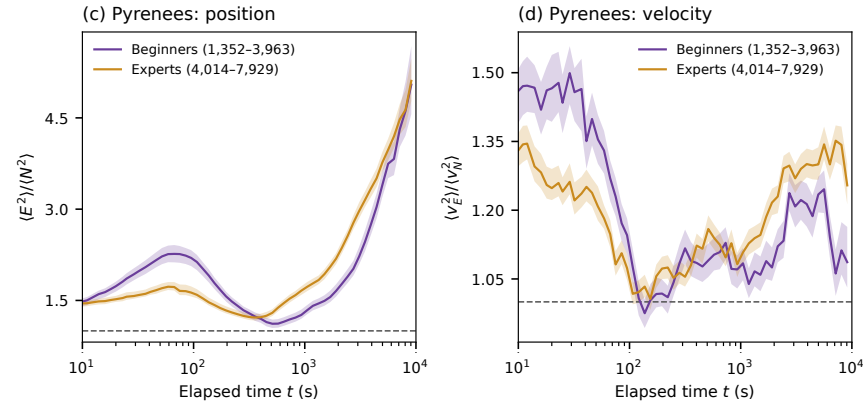
**Discussion.** Does the ordering persist after matching site, day, task and duration?

The fresh Alpine position and velocity ratios stay closer to unity for C/D/CCC over the supported grid. This is a descriptive ensemble observation. It does not establish individual isotropy, greater skill or a causal equipment effect. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.

**Pyrenees: the ordering of the two groups reverses**

**Discussion.** What changes between the early and later portions of the curves?

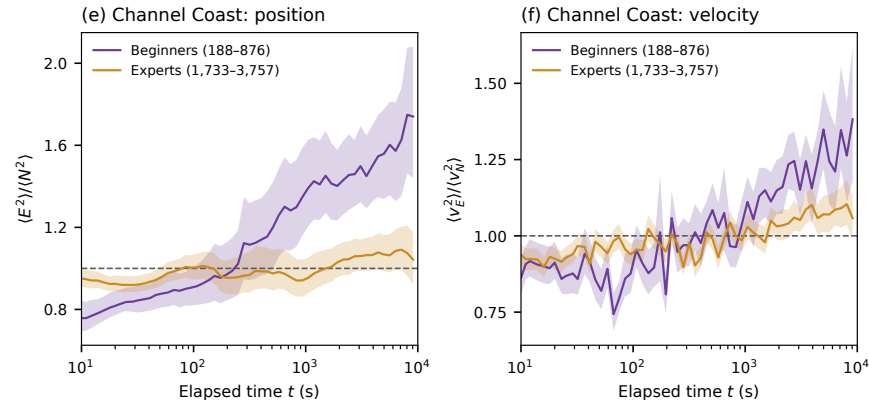
The Pyrenean curves do not support a uniform expert-proxy advantage. At late times both position ratios exceed five, while the velocity ordering also changes. Adjacent plotted times are correlated observations, not independent tests. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



**Channel Coast: the late position contrast is pronounced**

**Discussion.** Can matching conditions separate directional route choice from wind response?

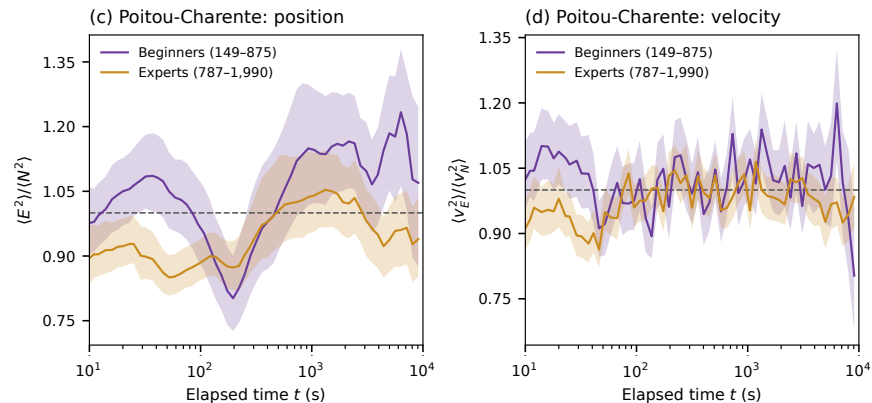
Near 9086 seconds the position ratios are about 1.74 for A/B and 1.04 for C/D/CCC, supported by 188 and 1733 paired flights. These are raw ensemble second moments and the samples differ strongly. Coastal cliffs and valleys prevent a wind-only interpretation. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



**Poitou-Charente: the group curves cross**

**Discussion.** Does this low-relief launch box reproduce the coastal observation?

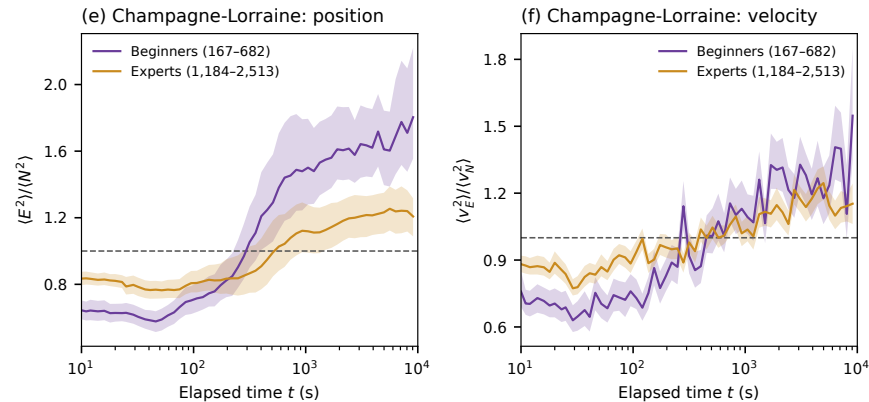
The crossing curves and broad overlapping bands do not support a general group ordering. These broad launch-location boxes are neither DEM-verified plains nor controls matched on weather. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



## Champagne-Lorraine: a late contrast does not keep C/D/CCC at unity

**Discussion.** Which parts of the coastal hypothesis transfer to this separate region?

A later position contrast appears, but the C/D/CCC ratio also rises above one. Region, route mixture, weather and declining support must be separated before claiming the same physical cause. Violet: A/B; ochre: C/D/CCC. Beginners and experts in the retained figure legend denote equipment proxies. Bands are pointwise 10–90-percent site-day bootstrap ranges, not intervals for the group difference.



## Hypothesis to discuss

Equipment and pilot choices may moderate directional constraints associated with wind and terrain. Similar mountain profiles may reflect shared route constraints.

## Evidence limiting that interpretation

The Pyrenean ordering reverses; class is an experience proxy; coastal terrain is not absent; late support differs; and ensemble balance does not establish individual isotropy or performance.

## The environmental interpretation remains a hypothesis

**Discussion.** What observation would distinguish this hypothesis from different sites, tasks, weather or route mixtures?

Preserve the proposed explanation as a useful hypothesis rather than silently removing it or presenting it as established. Similar curve shape does not by itself establish a common cause or an unavoidable terrain constraint. Wind and terrain can interact, so the coast/mountains split does not identify their separate effects.

$$\mathbf{v}_{\text{ground}} = \mathbf{v}_{\text{air}} + \mathbf{w}_h.$$

Upwind progress, air-relative performance, choice of route and geographic moment balance are different outcomes.

A first comparison would match overlapping site–day, duration and task support; then separate mean drift, centred variability and individual route orientation. Regional references show terrain alignment and a remaining coastal wind–PCA offset at flight altitude; local track-matched observations are still needed.

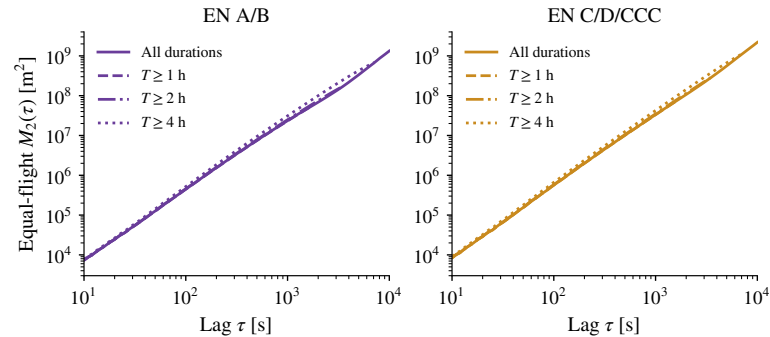
## Define the target before calling it resistance to wind

**Discussion.** Which target and matched comparison should be the first priority?

The site–day bootstrap adjusts one uncertainty structure; it does not perform matching or remove confounding. Where data permit, the same pilot on different equipment could help assess selection, but weather and task still require control. Differentiating velocity removes a constant wind term, but not its variation along the trajectory or the pilot’s manoeuvres.

Thesis §3.6

## Flight duration and equipment composition



For  $T \geq 4$  h: 7,415 A/B flights and 27,815 C/D/CCC flights.

Across 10–10,000 s, all-duration slopes are 1.728 and 1.788.

Equal-flight MSD; groups are nested. Threshold curves stop at  $\tau = T_0/2$ ; at least 30 flights per lag. All counts and common-range slopes.

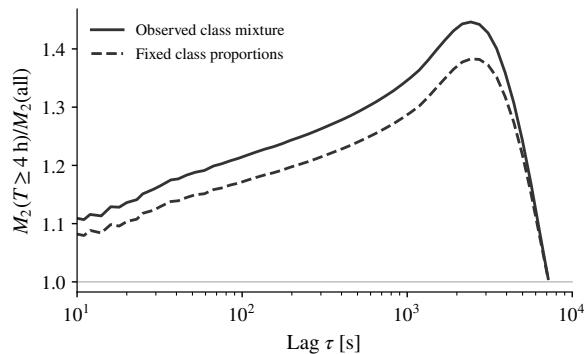
## Long flights occur in both equipment groups

The hypothesis that only experts contribute to flights longer than four hours is contradicted by the counts. Experts nevertheless become more frequent in the long-duration groups. On the common 10–1695 second interval, expert slopes exceed beginner slopes in every duration cell; within both classes slopes also increase between all flights and the four-hour group. This shorter interval is needed to compare all four nested cohorts under the half-duration rule. It is distinct from the main three-decade fit. Equipment, pilot selection, region and weather remain possible causes.

**Duration and equipment on the common 10–1695 s interval**

| Experience proxy | Duration     | Flights | $b$   | RMS residual (dex) |
|------------------|--------------|---------|-------|--------------------|
| Beginners        | All          | 54,292  | 1.755 | 0.022              |
| Beginners        | $T \geq 1$ h | 51,124  | 1.754 | 0.022              |
| Beginners        | $T \geq 2$ h | 31,768  | 1.767 | 0.020              |
| Beginners        | $T \geq 4$ h | 7,415   | 1.806 | 0.017              |
| Experts          | All          | 94,283  | 1.803 | 0.017              |
| Experts          | $T \geq 1$ h | 91,182  | 1.802 | 0.017              |
| Experts          | $T \geq 2$ h | 69,087  | 1.811 | 0.016              |
| Experts          | $T \geq 4$ h | 27,815  | 1.836 | 0.015              |

Counts are eligible flights, not independent windows; support falls with lag. RMS residual measures departures from a fitted line, not uncertainty. Threshold samples overlap.



$$M_{T_0}^{\text{fixed}}(\tau) = \sum_c \pi_c M_{T_0,c}(\tau).$$

Fixed weights:  $\pi_{A/B} = 0.365$ ,  
 $\pi_{C/D/CCC} = 0.635$ .

At  $\tau = 1695$  s the ratio falls from 1.42 to 1.35. A within-class contrast remains.

Both curves use the same two known equipment groups. Equality at one lag does not establish independence from flight duration.

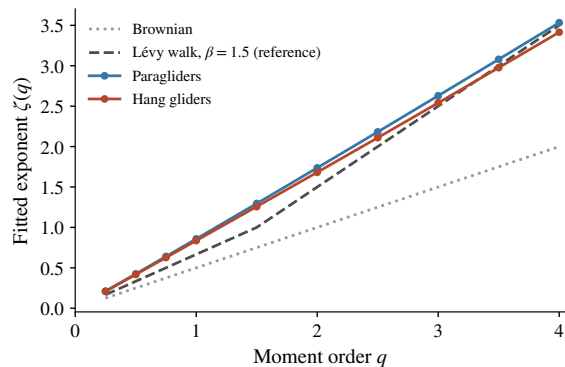
## Fixing equipment proportions removes part of the duration contrast

**Discussion.** Match site–day and task next. Convergence near 7200 s also reflects shorter flights leaving the reference.

The expert fraction among known classes rises from 63.5 percent overall to 78.9 percent in the four-hour cohort. Standardization uses the all-duration proportions at every lag for both numerator and denominator. It is a descriptive reweighting, not a causal adjustment. The remaining difference shows that the two-class proportions alone cannot account for the observed amplitude contrast. Near half the duration threshold, the reference loses shorter flights, so the curves are increasingly comparing similar populations.

Thesis §3.7.1

## The moment spectrum and the Lévy-walk benchmark



$$M_q = \langle R^q \rangle \propto \tau^{\zeta(q)}.$$

For a constant-speed renewal walk with isotropic independent headings and  $\psi(t) \sim t^{-1-\beta}$ ,  $1 < \beta < 2$ :

$$\zeta_{\text{LW}}(q) = \begin{cases} q/\beta, & q < \beta, \\ q + 1 - \beta, & q > \beta. \end{cases}$$

The empirical spectrum is nearly linear over  $q = 0.25$ –4.

Pooled windows.  $\beta = 1.5$  is illustrative, not a fitted null;  $q = \beta$  has a logarithmic correction. Zaburdaev, Denisov & Klafter (2015).

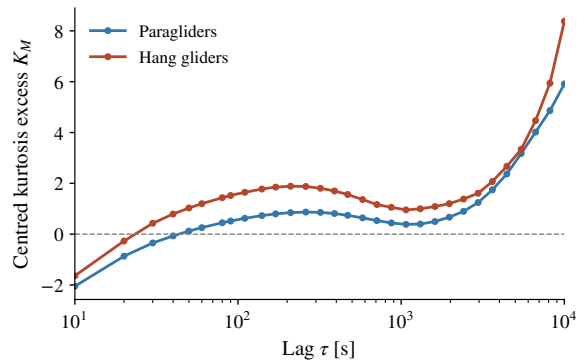
## The moment spectrum challenges a specific Lévy-walk benchmark

**Discussion.** Can finite-record simulations resolve the predicted branch change under our sampling and selection?

The current pooled moment spectrum has  $\zeta(2) = 1.737$  for paragliders and 1.682 for hang gliders. The corresponding  $\zeta(q)/q$  ranges are imported in the report. The illustrative beta=1.5 line is not a fitted null. A finite-record comparison must reproduce sample lengths, cadence, selection and weighting, and assess whether the asymptotic branch change would be resolvable.

Thesis §3.7.2

## Centred kurtosis and heterogeneity



Centred, full-covariance Mardia excess; pooled windows. Positive excess does not identify a power-law tail. Mardia (1970).

$$\mathbf{Z} = \Sigma^{-1/2}(\mathbf{X} - \boldsymbol{\mu}),$$

$$K_M = \langle |\mathbf{Z}|^4 \rangle - 8.$$

A nonsingular 2D Gaussian has  $K_M = 0$ , including anisotropic Gaussians.

The observed curve changes sign and rises at large lag, where support is weakest.

## Non-Gaussian pooled increments can arise from Gaussian mixtures

**Discussion.** Test Gaussianity within matched environments or phases before rejecting conditional Gaussian dynamics.

Explain the full covariance normalization: unequal east and north variances alone do not produce a nonzero population excess for a Gaussian. But mixing Gaussian variances does. If  $\mathbf{X}$  equals  $\sqrt{\mathbf{A}}$  times a fixed Gaussian vector,  $K$  equals eight times  $\text{Var}(\mathbf{A})$  divided by  $E(\mathbf{A})$  squared. This illustrates heterogeneity; it is not a correction formula for arbitrary flights. The finite-sample estimate need not vanish even for Gaussian data, and overlapping windows require a matched reference before significance claims.

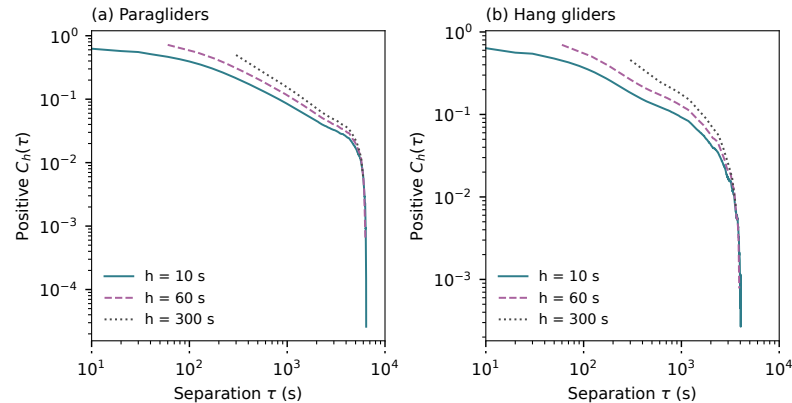
Thesis §3.7.3

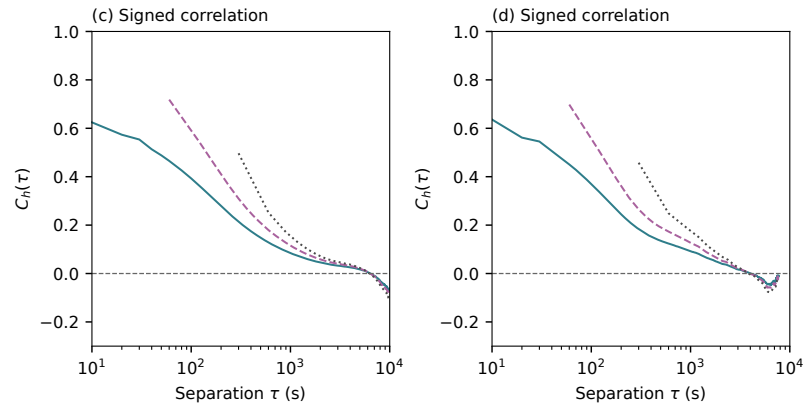
## Velocity memory across resolutions

**Velocity memory remains visible after averaging over minutes**

**Discussion.** Do the curves support one decay law over the displayed scales?

Use  $\mathbf{v}_h(t) = [\mathbf{r}(t+h) - \mathbf{r}(t)]/h$  for  $h=10, 60, 300$  seconds. Velocities use adjacent non-overlapping blocks; correlations are centred and normalized within flights before averaging. At 600 seconds the para values are 0.128, 0.180 and 0.252. These upper log panels show positive correlations only.



**Signed correlations retain the long-lag negative values**

**Discussion.** What share of the long-lag decrease could follow finite-record centring or route returns?

A log plot cannot show the negative part. Both disciplines have long-lag negative values in the signed representation. Coarse averaging can raise normalized correlations without increasing the underlying physical persistence time. At least twenty flights contribute to each plotted estimate.

**Finite velocity persistence gives a ballistic-to-diffusive crossover**

For the same stationary, centred velocity population,

$$M_{2,c}(\tau) = 2\sigma_v^2 \int_0^\tau (\tau - s)C(s) ds.$$

With  $C(s) = e^{-s/t_p}$ ,

$$M_{2,c} = 2\sigma_v^2 t_p [\tau - t_p(1 - e^{-\tau/t_p})].$$

Ballistic at short times, diffusive at long times.

Taylor (1922), pp. 207–208. This covariance identity cannot be applied directly to differently weighted heterogeneous means.

Thesis §3.7.4

## Model predictions and remaining tests

| Candidate                          | Prediction to test                 | Present conclusion                              |
|------------------------------------|------------------------------------|-------------------------------------------------|
| Homogeneous Brownian               | Linear MSD; independent increments | Challenged over the measured range.             |
| Constant drift + Brownian          | Drift-free $V_2 \propto \tau$      | Simple version challenged; mixtures unresolved. |
| Homogeneous fBm                    | Gaussian law; common $H_p$         | Challenged as one pooled law.                   |
| Exponential velocity memory        | Ballistic–diffusive crossover      | Remains a candidate.                            |
| Renewal Lévy walk, $1 < \beta < 2$ | Two-branch moment spectrum         | Finite-record comparison still needed.          |

“Challenged” means a descriptive mismatch. Calibrated statistical rejection is still to be done.

Each candidate needs its own parameters, environmental assumptions and observation protocol. A long-time diffusive limit has not been tested.

## Current constraints on candidate models

**Discussion.** Which models are challenged, and which remain open?

Ordinary Brownian motion is a poor descriptive model of the observed interval, but the asymptotic regime is unknown. fBm is challenged as a homogeneous pooled law; a phase-conditioned or environmentally heterogeneous Gaussian construction is not excluded. The exponential-persistence candidate deserves a joint MSD and VACF comparison. Standard Lévy walk needs finite-record calibration rather than rejection from an illustrative asymptotic line. An unbounded Lévy flight cannot literally describe continuous finite-speed flight; effective truncated versions need separate predictions.

Thesis §3.8

## Constraints for the phase analysis

$$M_q(\tau) = \sum_c w_c(\tau) M_q(\tau | c), \quad w_c = \frac{n_c}{\sum_{c'} n_{c'}}.$$

Here  $c$  partitions all admissible windows, including phase histories that cross boundaries. With equal flight weights, apply the identity within each flight before averaging.

The identity holds for any exhaustive labels. Recovering the global curve verifies accounting, not the physical accuracy of those labels.

### Phase-conditioned moments must include windows crossing transitions

**Discussion.** Which episode durations and between-phase dependencies could distinguish renewal dynamics from persistent motion?

Within-phase windows alone generally omit part of the global moment. Both the mixture weights and conditional moments depend on lag, so global exponents are not weighted averages of phase exponents. A glider moves while climbing; a climb is not automatically a waiting state.

1. **Define the model target.**

Ground-relative motion conditional on environment; residual motion requires an independently justified wind correction.

2. **Agree the first calibrated comparisons.**

Fit Brownian controls, fBm, finite persistence and a specified Lévy walk; simulate the same record lengths, selection and estimators.

3. **Choose the most informative next measurements.**

Site-day uncertainty and route matching now; phase durations, increments and between-phase dependence after segmentation validation.

We have measured constraints on growth, shape, geometry and memory. The generating process and an intrinsic Hurst exponent remain unresolved.

## Three decisions for this meeting

**Discussion.** Ground-relative motion conditional on environment, or a residual after an independently justified wind correction?

Agree the physical target, a small set of explicit candidate processes, and a sampling model for uncertainty. No rotation or rescaling has recovered still-air motion. Phase-conditioned comparisons should include transitions and between-episode dependence. Calibrated comparisons can begin before declaring a final physical model.

$$Q_{Y^2}(p) = Q_Y(p)^2, \quad H_{Y^2,p} = 2H_{Y,p} \quad \text{for } Y \geq 0.$$

Squaring bin edges preserves their probability masses. A monotone transform also preserves the supremum distance between the correspondingly transformed CDFs.

If  $f_Y(0^+) = c > 0$ , then  $f_{Y^2}(s) \sim c/(2\sqrt{s})$  near zero. This Jacobian effect says nothing about a heavy upper tail.

## Squared laws transform the same distributional evidence

**Discussion.** Which representation communicates the same result more clearly: displacement magnitude or its square?

The appendix includes the squared-law figures for both disciplines. Replotting the same observations after a monotone transformation is useful for connecting quantiles with second moments; it does not multiply the amount of independent evidence. Zero mass and dimensional units still need to be carried consistently.

Sources and the scope of the evidence

|                                                                      |                                  |
|----------------------------------------------------------------------|----------------------------------|
| Zaburdaev, Denisov & Klafter (2015), <i>Lévy walks</i>               | Moment-spectrum benchmark        |
| Mandelbrot & Van Ness (1968), <i>Fractional Brownian motions</i>     | fBm construction                 |
| Taylor (1922), <i>Diffusion by continuous movements</i>              | Velocity / displacement relation |
| Mardia (1970), <i>Measures of multivariate skewness and kurtosis</i> | Centred kurtosis                 |
| Efron (1979), <i>Bootstrap methods</i>                               | Resampling principle             |

Numerical evidence: current Chapter 3 reports, frozen in the accompanying data and asset manifests. Bibliographic access limitations are retained in the repository audit; citations do not validate the chosen sampling design or constitute calibrated rejection tests.

**Joint laws and environmental interpretation: additional sources**

Fraiman et al., Cramér–Wold results

Joint laws and projections

Didier & Pipiras, operator fractional Brownian motions

Beyond scalar rescaling

DHV classification guidance

Class is not a performance measure

DREAL Normandie, Le Pays de Caux

Coastal cliffs and valleys

The cited pages support specific mathematical or observational cautions. They do not identify the causes of the regional contrasts measured here.